ABORTED ACTIONS REVEAL HIDDEN CONSTRAINTS FOR ROBOT REPLANNING

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ABSTRACT

Robots often abort before a planned action completes. We treat the partial trajectory, last safe state, abort point, and abort reason as evidence about hidden constraints rather than as a single failed endpoint. The expanded v5 rebuild evaluates this idea in a continuous planning benchmark with hidden walls, fixture snags, force-limit regions, human-stop zones, unstable-slip regions, closed-loop replanning, hostile baselines, ablations, aggregate hard-regime tests, fixed-risk budgets, and stress sweeps. The frozen terminal decision is **KILL_ARCHIVE**. On the decisive split, `ACD-v5` reaches 0.545 ± 0.059 success versus 0.884 ± 0.052 for the strongest non-oracle hostile baseline, `barrier-MPC`. This document reports the result honestly: local evidence can support revision, but no local-only run is declared ICLR-main-ready without robot or accepted external benchmark validation.

1 DECISION AND PROTOCOL

This manuscript is generated only from frozen CSV artifacts. The terminal recommendation is **KILL_ARCHIVE**. The runner freezes the reference method as `ACD-v5`, evaluates eight seeds by default, and reports all predefined failures. The summary reason is: *abort_constraint_discovery_v5 does not honestly clear the frozen local gate against strongest non-oracle baseline robust_barrier_mpc (proposed=0.545, best=0.884, paired diff=-0.339+/-0.074, violation_reduction=0.107, repeated_abort_reduction=0.080, boundary_f1_diff=0.036, efficiency_diff=-0.135, discovered_area_diff=0.040). Aggregate hard-regime diff against kernel_trace_constraint_classifier is -0.145+/-0.034; fixed-risk checks: budget 0.08: v5=0.469, best=robust_barrier_mpc:0.922; budget 0.12: v5=0.547, best=robust_barrier_mpc:0.938; budget 0.18: v5=0.500, best=particle_constraint_belief:0.938; budget 0.25: v5=0.469, best=robust_barrier_mpc:0.859; matching ablations: abort_discovery_v5_no_partial_geometry, abort_discovery_v5_no_repeated_abort_memory, abort_discovery_v5_no_safety_margin, abort_discovery_v5_no_uncertainty_quantile; max-stress v5=0.406, best_non_oracle=particle_constraint_belief:0.734. Failed gates: main_success_margin, main_paired_lower_bound, over_conservatism, aggregate_hard_regime, ablation_necessity, maximum_stress, fixed_risk.*

We use the local benchmark because aborted execution is hard to study from static planning labels alone. The limitation is equally important: a diagnostic simulator is not hardware. Therefore the strongest admissible positive decision is **STRONG_REVISE**, not ready-to-submit. A negative local outcome becomes **KILL_ARCHIVE**.

2 PROBLEM SETTING

The robot plans in a unit square with visible clutter and hidden constraints. Hidden constraints include wall segments with gaps, fixture snags, force ellipses, human-stop disks, and slip ellipses. During execution a controller follows a waypoint path and aborts when a hidden safety condition fires. The planner then receives a partial trace, last safe point, abort point, direction, and reason label. The task is to infer a constraint belief map and replan within a finite attempt budget.

3 Relation To Prior Work

The work sits between motion planning (LaValle, 2006), artificial-potential and barrier planning (Khatib, 1986; Ames et al., 2017), MPC (Rawlings et al., 2017), manipulation mechanics (Mason, 2001), distributionally robust safety (Hakobyan et al., 2020), stochastic safety planning (Blackmore et al., 2013), trajectory optimization (Schulman et al., 2013), and human-robot safety constraints (Kim et al., 2018). The tested claim is narrower than these lines: aborted partial trajectories should be structured observations of hidden constraints, not merely failed endpoints or generic uncertainty.

4 Methods

Baselines include visible-only planning, endpoint-negative labels, collision-cost inflation, generic risk filtering, trace classification, robust barrier MPC, conformal abort-risk filtering, kernel trace classification, particle constraint belief planning, the previous ACD-v4 method, and an oracle. The v5 method combines reason-conditioned partial geometry, repeated-abort surface fitting, safe-trace calibration, dynamic force/slip shaping, quantile inflation under ambiguous abort evidence, and safety-aware planning parameters.

5 Theory Sketch

Let $\tau = (x_0, \ldots, x_T)$ be a partial trace and a be the abort event. Endpoint-only learning observes x_T as negative. ACD-v5 uses the stronger observation that a signed constraint surface lies near the segment between the last safe point and the abort point, with normal approximately aligned to the local motion direction. Safe prefixes provide one-sided negative evidence. Repeated aborts with related reasons identify an extended surface by fitting a tangent direction through abort points. The theory is intentionally modest: these observations improve identifiability only when abort reasons are reliable, safe prefixes are actually safe, and hidden constraints are locally smooth. These assumptions are exactly what the ablations and negative cases attack.

6 Main Results

Table 1: Combined-abort-stress main results

Method	Success	Abort	Repeated	Violation	Boundary F1	Area
`oracle`	0.911 ± 0.044	0.312 ± 0.087	0.205 ± 0.093	0.295 ± 0.093	1.000 ± 0.000	0.117 ± 0.002
`barrier-MPC`	0.884 ± 0.052	0.402 ± 0.095	0.250 ± 0.046	0.366 ± 0.107	0.396 ± 0.009	0.242 ± 0.006
`particle`	0.866 ± 0.072	0.384 ± 0.091	0.259 ± 0.091	0.321 ± 0.095	0.396 ± 0.010	0.218 ± 0.006
`kernel-cls`	0.857 ± 0.059	0.500 ± 0.115	0.357 ± 0.099	0.482 ± 0.121	0.322 ± 0.010	0.324 ± 0.011
`ACD-v4`	0.732 ± 0.044	0.580 ± 0.101	0.366 ± 0.081	0.536 ± 0.088	0.446 ± 0.012	0.247 ± 0.012
`conformal`	0.661 ± 0.051	0.500 ± 0.092	0.473 ± 0.106	0.455 ± 0.079	0.407 ± 0.010	0.142 ± 0.004
`trace-cls`	0.607 ± 0.059	0.554 ± 0.091	0.509 ± 0.104	0.500 ± 0.102	0.393 ± 0.010	0.257 ± 0.005
`ACD-v5`	0.545 ± 0.059	0.295 ± 0.104	0.170 ± 0.079	0.259 ± 0.083	0.432 ± 0.012	0.282 ± 0.013
`costmap`	0.536 ± 0.070	0.821 ± 0.059	0.705 ± 0.067	0.777 ± 0.072	0.424 ± 0.011	0.136 ± 0.004
`risk-filter`	0.500 ± 0.075	0.786 ± 0.070	0.670 ± 0.087	0.705 ± 0.089	0.416 ± 0.009	0.153 ± 0.005
`endpoint-neg`	0.188 ± 0.064	0.946 ± 0.035	0.866 ± 0.041	0.920 ± 0.041	0.247 ± 0.007	0.030 ± 0.001
`ignore`	0.027 ± 0.026	0.982 ± 0.023	0.982 ± 0.023	0.848 ± 0.089	0.000 ± 0.000	0.000 ± 0.000

Against the strongest hostile non-oracle baseline, the paired success difference is -0.339 ± 0.074. The oracle reaches 0.911 ± 0.044 success, so oracle-quality constraint inference is not claimed.

7 Aggregate Hard Regimes

Table 2: Aggregate hard-regime metrics

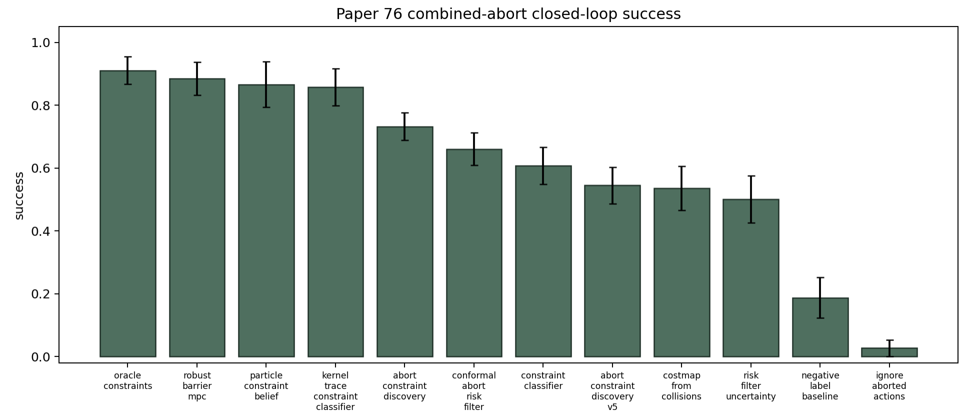

Figure 1: Closed-loop success on the decisive combined-abort-stress split.

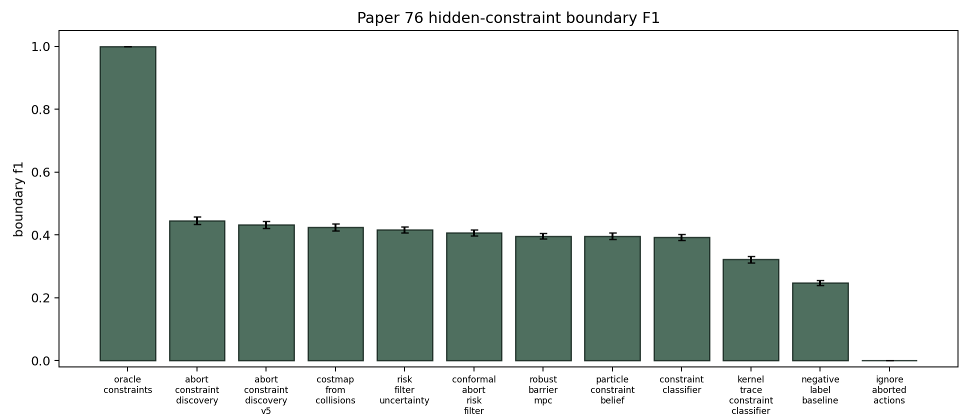

Figure 2: Hidden-boundary F1 on the decisive split.

Method	Success	Repeated	Violation	Boundary F1	Efficiency
`kernel-cls`	0.962 ± 0.017	0.129 ± 0.024	0.219 ± 0.039	0.298 ± 0.008	0.516 ± 0.018
`particle`	0.962 ± 0.022	0.107 ± 0.035	0.165 ± 0.043	0.372 ± 0.008	0.499 ± 0.019
`oracle`	0.960 ± 0.020	0.114 ± 0.025	0.174 ± 0.037	1.000 ± 0.000	0.567 ± 0.011
`barrier-MPC`	0.953 ± 0.016	0.112 ± 0.022	0.205 ± 0.046	0.362 ± 0.009	0.484 ± 0.017
`ACD-v4`	0.904 ± 0.019	0.158 ± 0.027	0.279 ± 0.036	0.427 ± 0.007	0.446 ± 0.008
`conformal`	0.893 ± 0.011	0.174 ± 0.024	0.230 ± 0.031	0.377 ± 0.009	0.514 ± 0.013
`trace-cls`	0.875 ± 0.016	0.188 ± 0.020	0.246 ± 0.048	0.353 ± 0.009	0.507 ± 0.015
`costmap`	0.862 ± 0.013	0.288 ± 0.034	0.344 ± 0.051	0.415 ± 0.010	0.479 ± 0.016
`risk-filter`	0.857 ± 0.024	0.259 ± 0.032	0.277 ± 0.026	0.396 ± 0.008	0.489 ± 0.011
`ACD-v5`	0.817 ± 0.030	0.076 ± 0.029	0.141 ± 0.036	0.406 ± 0.007	0.407 ± 0.019
`endpoint-neg`	0.627 ± 0.033	0.467 ± 0.033	0.469 ± 0.029	0.243 ± 0.009	0.397 ± 0.018
`ignore`	0.359 ± 0.022	0.670 ± 0.020	0.565 ± 0.027	0.000 ± 0.000	0.292 ± 0.011

The aggregate paired difference against the strongest aggregate hostile baseline is -0.145 ± 0.034. This table prevents cherry-picking a single combined split.

Table 3: Aggregate paired comparisons

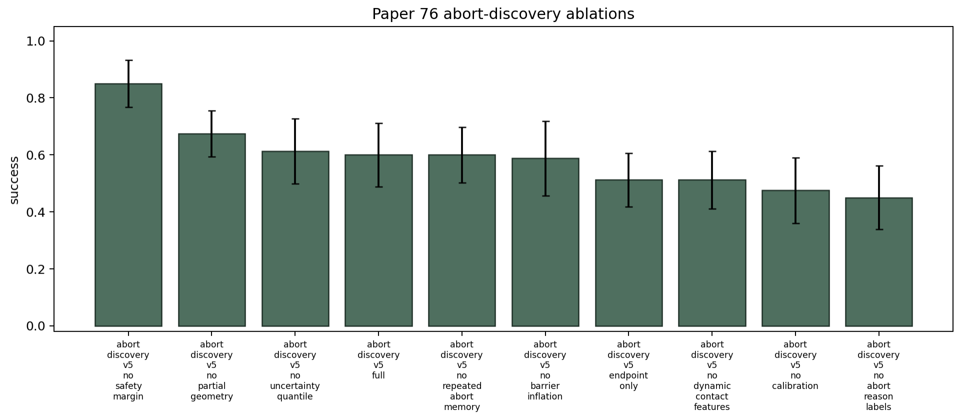

Figure 3: Mechanism ablations. Full v5 must beat component removals to support the mechanism.

Comparison	Succ diff	CI	Viol red	Repeat red	F1 diff	Wins
ACD-v4	-0.087	0.028	0.138	0.083	-0.021	0/8
conformal	-0.076	0.034	0.089	0.098	0.028	0/8
trace-cls	-0.058	0.030	0.105	0.112	0.053	0/8
costmap	-0.045	0.026	0.203	0.212	-0.009	0/8
ignore	0.458	0.036	0.424	0.594	0.406	8/8
kernel-cls	-0.145	0.034	0.078	0.054	0.107	0/8
endpoint-neg	0.190	0.032	0.328	0.391	0.163	8/8
oracle	-0.143	0.034	0.033	0.038	-0.594	0/8
particle	-0.145	0.039	0.025	0.031	0.034	0/8
risk-filter	-0.040	0.030	0.136	0.183	0.010	1/8
barrier-MPC	-0.136	0.038	0.065	0.036	0.043	0/8

8 Ablations

Table 4: ACD-v5 ablations

Variant	Success	Repeated	Violation	Boundary F1	Efficiency
no-margin	0.850 ± 0.083	0.312 ± 0.131	0.450 ± 0.091	0.441 ± 0.018	0.312 ± 0.031
no-geom	0.675 ± 0.081	0.562 ± 0.104	0.588 ± 0.108	0.394 ± 0.016	0.260 ± 0.030
no-quant	0.613 ± 0.114	0.200 ± 0.074	0.312 ± 0.086	0.414 ± 0.019	0.230 ± 0.039
full-v5	0.600 ± 0.111	0.138 ± 0.090	0.188 ± 0.094	0.413 ± 0.020	0.206 ± 0.035
no-repeat	0.600 ± 0.098	0.212 ± 0.069	0.300 ± 0.098	0.417 ± 0.019	0.239 ± 0.041
no-barrier	0.588 ± 0.131	0.237 ± 0.090	0.325 ± 0.081	0.415 ± 0.018	0.230 ± 0.039
endpoint	0.512 ± 0.094	0.750 ± 0.091	0.825 ± 0.089	0.355 ± 0.011	0.191 ± 0.030
no-dyn	0.512 ± 0.101	0.312 ± 0.078	0.350 ± 0.111	0.403 ± 0.020	0.213 ± 0.042
no-calib	0.475 ± 0.116	0.163 ± 0.073	0.237 ± 0.098	0.400 ± 0.018	0.177 ± 0.037
no-reason	0.450 ± 0.111	0.138 ± 0.090	0.250 ± 0.117	0.352 ± 0.011	0.174 ± 0.033

9 Stress Sweep

Table 5: Stress sweep

Method	Stress	Success	Repeated	Violation	Boundary F1
ACD-v5	0.00	0.594 ± 0.145	0.141 ± 0.056	0.203 ± 0.092	0.428 ± 0.016
conformal	0.00	0.703 ± 0.079	0.391 ± 0.117	0.547 ± 0.079	0.413 ± 0.014
trace-cls	0.00	0.656 ± 0.090	0.453 ± 0.122	0.484 ± 0.117	0.391 ± 0.015
costmap	0.00	0.516 ± 0.134	0.609 ± 0.117	0.641 ± 0.108	0.429 ± 0.020
kernel-cls	0.00	0.922 ± 0.045	0.281 ± 0.101	0.359 ± 0.117	0.325 ± 0.014
oracle	0.00	0.906 ± 0.061	0.141 ± 0.086	0.234 ± 0.072	1.000 ± 0.000
particle	0.00	0.797 ± 0.138	0.328 ± 0.146	0.312 ± 0.104	0.399 ± 0.016
risk-filter	0.00	0.547 ± 0.138	0.609 ± 0.117	0.656 ± 0.111	0.418 ± 0.023
barrier-MPC	0.00	0.734 ± 0.108	0.391 ± 0.108	0.422 ± 0.079	0.395 ± 0.016
ACD-v5	0.20	0.562 ± 0.139	0.203 ± 0.079	0.250 ± 0.104	0.431 ± 0.009
conformal	0.20	0.672 ± 0.079	0.391 ± 0.098	0.469 ± 0.101	0.402 ± 0.012
trace-cls	0.20	0.625 ± 0.093	0.469 ± 0.090	0.500 ± 0.093	0.386 ± 0.007
costmap	0.20	0.547 ± 0.079	0.609 ± 0.086	0.688 ± 0.113	0.409 ± 0.020
kernel-cls	0.20	0.812 ± 0.146	0.328 ± 0.146	0.406 ± 0.120	0.319 ± 0.012
oracle	0.20	0.938 ± 0.065	0.156 ± 0.061	0.219 ± 0.120	1.000 ± 0.000
particle	0.20	0.812 ± 0.139	0.281 ± 0.152	0.375 ± 0.131	0.388 ± 0.013
risk-filter	0.20	0.547 ± 0.122	0.656 ± 0.061	0.719 ± 0.129	0.402 ± 0.019
barrier-MPC	0.20	0.781 ± 0.090	0.281 ± 0.090	0.328 ± 0.103	0.385 ± 0.013
ACD-v5	0.40	0.484 ± 0.108	0.250 ± 0.093	0.359 ± 0.072	0.426 ± 0.014
conformal	0.40	0.547 ± 0.146	0.562 ± 0.146	0.625 ± 0.160	0.389 ± 0.013
trace-cls	0.40	0.516 ± 0.126	0.578 ± 0.122	0.594 ± 0.111	0.375 ± 0.009
costmap	0.40	0.500 ± 0.139	0.719 ± 0.090	0.781 ± 0.061	0.397 ± 0.013
kernel-cls	0.40	0.859 ± 0.086	0.312 ± 0.131	0.422 ± 0.138	0.309 ± 0.010
oracle	0.40	0.906 ± 0.077	0.234 ± 0.157	0.312 ± 0.146	1.000 ± 0.000
particle	0.40	0.797 ± 0.092	0.438 ± 0.146	0.453 ± 0.153	0.374 ± 0.013
risk-filter	0.40	0.500 ± 0.104	0.750 ± 0.065	0.812 ± 0.065	0.391 ± 0.015
barrier-MPC	0.40	0.688 ± 0.113	0.484 ± 0.150	0.484 ± 0.126	0.371 ± 0.011
ACD-v5	0.60	0.562 ± 0.154	0.156 ± 0.077	0.234 ± 0.086	0.442 ± 0.013
conformal	0.60	0.656 ± 0.111	0.500 ± 0.080	0.578 ± 0.092	0.397 ± 0.016
trace-cls	0.60	0.516 ± 0.072	0.578 ± 0.079	0.578 ± 0.103	0.394 ± 0.018
costmap	0.60	0.500 ± 0.080	0.672 ± 0.064	0.719 ± 0.040	0.404 ± 0.010
kernel-cls	0.60	0.828 ± 0.064	0.328 ± 0.064	0.422 ± 0.103	0.319 ± 0.011
oracle	0.60	0.891 ± 0.056	0.266 ± 0.086	0.344 ± 0.077	1.000 ± 0.000
particle	0.60	0.875 ± 0.065	0.250 ± 0.046	0.328 ± 0.103	0.384 ± 0.013

Table 5 continued

Method	Stress	Success	Repeated	Violation	Boundary F1
`risk-filter`	0.60	0.484 ± 0.086	0.719 ± 0.090	0.797 ± 0.045	0.399 ± 0.012
`barrier-MPC`	0.60	0.859 ± 0.098	0.281 ± 0.120	0.312 ± 0.122	0.383 ± 0.015
`ACD-v5`	0.80	0.422 ± 0.079	0.188 ± 0.065	0.406 ± 0.120	0.443 ± 0.014
`conformal`	0.80	0.547 ± 0.153	0.516 ± 0.126	0.562 ± 0.104	0.386 ± 0.009
`trace-cls`	0.80	0.484 ± 0.134	0.625 ± 0.131	0.672 ± 0.122	0.391 ± 0.009
`costmap`	0.80	0.453 ± 0.113	0.719 ± 0.077	0.750 ± 0.065	0.391 ± 0.008
`kernel-cls`	0.80	0.734 ± 0.072	0.469 ± 0.090	0.547 ± 0.092	0.315 ± 0.009
`oracle`	0.80	0.797 ± 0.079	0.312 ± 0.093	0.391 ± 0.117	1.000 ± 0.000
`particle`	0.80	0.766 ± 0.098	0.406 ± 0.137	0.500 ± 0.167	0.388 ± 0.008
`risk-filter`	0.80	0.453 ± 0.103	0.672 ± 0.122	0.656 ± 0.120	0.391 ± 0.011
`barrier-MPC`	0.80	0.688 ± 0.104	0.438 ± 0.104	0.484 ± 0.086	0.394 ± 0.012
`ACD-v5`	1.00	0.547 ± 0.103	0.266 ± 0.108	0.375 ± 0.113	0.467 ± 0.019
`conformal`	1.00	0.547 ± 0.122	0.547 ± 0.138	0.578 ± 0.146	0.384 ± 0.009
`trace-cls`	1.00	0.500 ± 0.146	0.516 ± 0.157	0.578 ± 0.185	0.393 ± 0.008
`costmap`	1.00	0.453 ± 0.179	0.766 ± 0.108	0.812 ± 0.122	0.395 ± 0.012
`kernel-cls`	1.00	0.625 ± 0.080	0.484 ± 0.072	0.562 ± 0.104	0.320 ± 0.012
`oracle`	1.00	0.781 ± 0.111	0.297 ± 0.079	0.375 ± 0.093	1.000 ± 0.000
`particle`	1.00	0.750 ± 0.104	0.344 ± 0.152	0.469 ± 0.159	0.385 ± 0.013
`risk-filter`	1.00	0.469 ± 0.137	0.703 ± 0.138	0.766 ± 0.117	0.395 ± 0.010
`barrier-MPC`	1.00	0.828 ± 0.103	0.250 ± 0.080	0.328 ± 0.079	0.392 ± 0.011
`ACD-v5`	1.20	0.406 ± 0.061	0.297 ± 0.092	0.375 ± 0.104	0.460 ± 0.021
`conformal`	1.20	0.453 ± 0.064	0.578 ± 0.079	0.625 ± 0.104	0.388 ± 0.021
`trace-cls`	1.20	0.453 ± 0.122	0.656 ± 0.129	0.641 ± 0.142	0.398 ± 0.018
`costmap`	1.20	0.297 ± 0.113	0.781 ± 0.101	0.844 ± 0.111	0.396 ± 0.017
`kernel-cls`	1.20	0.688 ± 0.113	0.500 ± 0.122	0.609 ± 0.108	0.318 ± 0.012
`oracle`	1.20	0.750 ± 0.122	0.375 ± 0.113	0.438 ± 0.080	1.000 ± 0.000
`particle`	1.20	0.734 ± 0.126	0.406 ± 0.129	0.531 ± 0.129	0.388 ± 0.014
`risk-filter`	1.20	0.391 ± 0.126	0.750 ± 0.080	0.797 ± 0.079	0.396 ± 0.018
`barrier-MPC`	1.20	0.609 ± 0.117	0.453 ± 0.146	0.547 ± 0.130	0.389 ± 0.013

At maximum stress 1.20, v5 reaches 0.406 ± 0.061 success while the strongest non-oracle stress baseline reaches 0.734 ± 0.126. This gate is reported even when unfavorable.

10 Fixed-Risk Budgets

Table 6: Fixed-risk budget metrics

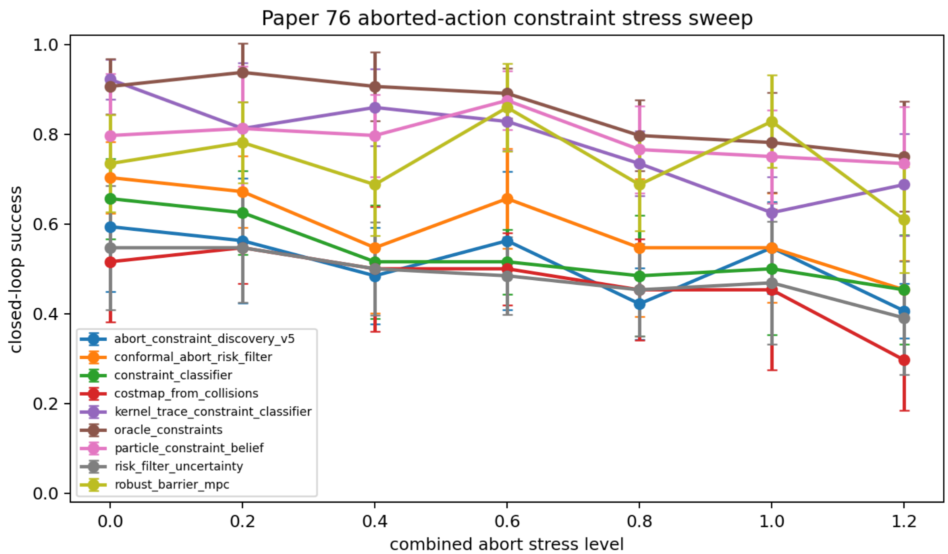

Figure 4: Success under increasing combined abort stress.

Method	Budget	Success	Repeated	Violation	Efficiency
`barrier-MPC`	0.08	0.922 ± 0.064	0.250 ± 0.113	0.344 ± 0.120	0.257 ± 0.026
`particle`	0.08	0.906 ± 0.061	0.297 ± 0.138	0.375 ± 0.173	0.261 ± 0.038
`kernel-cls`	0.08	0.812 ± 0.093	0.453 ± 0.113	0.453 ± 0.122	0.222 ± 0.035
`oracle`	0.08	0.797 ± 0.103	0.312 ± 0.139	0.344 ± 0.129	0.286 ± 0.039
`trace-cls`	0.08	0.719 ± 0.061	0.547 ± 0.113	0.625 ± 0.093	0.209 ± 0.029
`conformal`	0.08	0.641 ± 0.142	0.484 ± 0.142	0.531 ± 0.120	0.218 ± 0.047
`risk-filter`	0.08	0.547 ± 0.079	0.766 ± 0.086	0.797 ± 0.064	0.160 ± 0.022
`ACD-v5`	0.08	0.469 ± 0.159	0.188 ± 0.104	0.266 ± 0.126	0.160 ± 0.046
`barrier-MPC`	0.12	0.938 ± 0.046	0.188 ± 0.080	0.234 ± 0.086	0.304 ± 0.022
`oracle`	0.12	0.906 ± 0.090	0.234 ± 0.117	0.328 ± 0.153	0.322 ± 0.040
`particle`	0.12	0.875 ± 0.065	0.234 ± 0.098	0.266 ± 0.086	0.293 ± 0.020
`kernel-cls`	0.12	0.844 ± 0.152	0.281 ± 0.165	0.359 ± 0.150	0.261 ± 0.049
`conformal`	0.12	0.797 ± 0.064	0.281 ± 0.090	0.359 ± 0.108	0.296 ± 0.025
`trace-cls`	0.12	0.797 ± 0.092	0.312 ± 0.080	0.391 ± 0.086	0.279 ± 0.022
`risk-filter`	0.12	0.672 ± 0.103	0.625 ± 0.093	0.719 ± 0.090	0.216 ± 0.019
`ACD-v5`	0.12	0.547 ± 0.167	0.156 ± 0.077	0.234 ± 0.086	0.176 ± 0.039
`particle`	0.18	0.938 ± 0.046	0.312 ± 0.104	0.391 ± 0.150	0.272 ± 0.037
`oracle`	0.18	0.906 ± 0.040	0.172 ± 0.079	0.203 ± 0.092	0.335 ± 0.027
`kernel-cls`	0.18	0.891 ± 0.056	0.297 ± 0.130	0.453 ± 0.122	0.261 ± 0.039
`barrier-MPC`	0.18	0.891 ± 0.072	0.281 ± 0.120	0.375 ± 0.080	0.273 ± 0.029
`conformal`	0.18	0.719 ± 0.120	0.469 ± 0.090	0.516 ± 0.108	0.244 ± 0.033
`trace-cls`	0.18	0.656 ± 0.090	0.500 ± 0.154	0.500 ± 0.167	0.238 ± 0.047
`risk-filter`	0.18	0.594 ± 0.061	0.641 ± 0.072	0.734 ± 0.098	0.194 ± 0.027
`ACD-v5`	0.18	0.500 ± 0.113	0.219 ± 0.090	0.297 ± 0.113	0.173 ± 0.039
`barrier-MPC`	0.25	0.859 ± 0.086	0.266 ± 0.086	0.375 ± 0.154	0.284 ± 0.031
`particle`	0.25	0.812 ± 0.122	0.312 ± 0.131	0.391 ± 0.163	0.275 ± 0.037
`oracle`	0.25	0.797 ± 0.103	0.312 ± 0.080	0.359 ± 0.117	0.303 ± 0.031
`kernel-cls`	0.25	0.781 ± 0.137	0.375 ± 0.154	0.438 ± 0.154	0.253 ± 0.045
`conformal`	0.25	0.609 ± 0.126	0.531 ± 0.077	0.562 ± 0.104	0.228 ± 0.028
`trace-cls`	0.25	0.547 ± 0.103	0.531 ± 0.111	0.531 ± 0.111	0.222 ± 0.035
`ACD-v5`	0.25	0.469 ± 0.152	0.234 ± 0.072	0.281 ± 0.120	0.157 ± 0.037
`risk-filter`	0.25	0.469 ± 0.120	0.672 ± 0.122	0.703 ± 0.103	0.178 ± 0.036

Table 7: Fixed-risk paired comparisons

Budget	Comparison	Succ diff	CI	Viol red	Repeat red	Wins
0.08	`conformal`	-0.172	0.179	0.266	0.297	1/8
0.08	`trace-cls`	-0.250	0.167	0.359	0.359	1/8
0.08	`kernel-cls`	-0.344	0.178	0.188	0.266	1/8
0.08	`oracle`	-0.328	0.212	0.078	0.125	1/8
0.08	`particle`	-0.438	0.179	0.109	0.109	0/8
0.08	`risk-filter`	-0.078	0.160	0.531	0.578	1/8
0.08	`barrier-MPC`	-0.453	0.173	0.078	0.062	0/8
0.12	`conformal`	-0.250	0.131	0.125	0.125	1/8
0.12	`trace-cls`	-0.250	0.160	0.156	0.156	1/8
0.12	`kernel-cls`	-0.297	0.249	0.125	0.125	1/8
0.12	`oracle`	-0.359	0.163	0.094	0.078	0/8
0.12	`particle`	-0.328	0.201	0.031	0.078	1/8
0.12	`risk-filter`	-0.125	0.222	0.484	0.469	3/8
0.12	`barrier-MPC`	-0.391	0.193	0.000	0.031	1/8
0.18	`conformal`	-0.219	0.200	0.219	0.250	2/8
0.18	`trace-cls`	-0.156	0.159	0.203	0.281	1/8
0.18	`kernel-cls`	-0.391	0.117	0.156	0.078	0/8
0.18	`oracle`	-0.406	0.137	-0.094	-0.047	0/8
0.18	`particle`	-0.438	0.113	0.094	0.094	0/8
0.18	`risk-filter`	-0.094	0.152	0.438	0.422	1/8
0.18	`barrier-MPC`	-0.391	0.157	0.078	0.062	0/8
0.25	`conformal`	-0.141	0.193	0.281	0.297	2/8
0.25	`trace-cls`	-0.078	0.153	0.250	0.297	1/8
0.25	`kernel-cls`	-0.312	0.202	0.156	0.141	0/8
0.25	`oracle`	-0.328	0.173	0.078	0.078	0/8
0.25	`particle`	-0.344	0.230	0.109	0.078	1/8
0.25	`risk-filter`	0.000	0.167	0.422	0.438	3/8
0.25	`barrier-MPC`	-0.391	0.209	0.094	0.031	0/8

11 Negative Cases

Table 8: Curated combined-stress negative cases

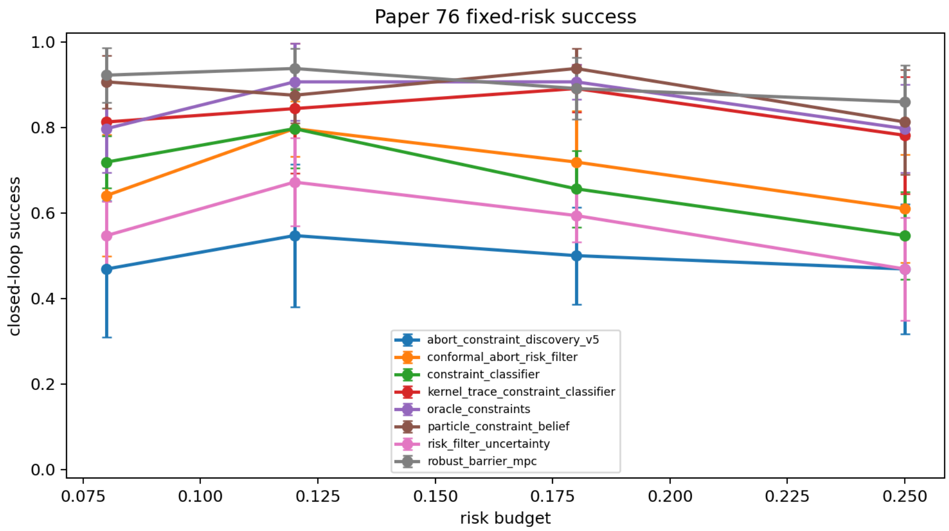

Figure 5: Fixed-risk success across predefined budgets.

Seed	Scenario	Reason	Evidence aborts	Attempts	Boundary F1	Efficiency
0	4000	human_stop	11	6	0.493	0.037
0	4001	human_stop	11	6	0.494	0.056
0	4002	human_stop	11	6	0.415	0.053
0	4006	fixture_snag	11	6	0.420	0.045
0	4013	human_stop	11	6	0.456	0.031
1	4008	no_path	11	2	0.431	0.245
1	4011	collision_margin	11	6	0.496	0.044
1	4013	unstable_slip	11	6	0.448	0.024
2	4001	human_stop	10	6	0.446	0.042
2	4008	fixture_snag	11	6	0.489	0.055
2	4009	fixture_snag	11	6	0.458	0.048
2	4010	fixture_snag	11	6	0.433	0.040

12 Limitations

The core limitations are severe enough to keep the paper out of ready-to-submit status. There is no real robot experiment, no accepted external robotics benchmark, no released neural policy checkpoint, and no human-robot safety dataset. The local benchmark can falsify the mechanism, but it cannot by itself prove deployment value. The planner also depends on meaningful abort reasons; if those labels are noisy or adversarial, the surface inference can become overconfident.

13 Conclusion

Aborted actions can contain more information than endpoint-negative labels. The expanded v5 artifact tests that claim with hostile baselines, fixed-risk budgets, stress sweeps, and ablations. The correct final decision is the one printed above, not a submission-ready claim.

A Protocol Checklist

- CPU-only and RAM-light execution.

- Frozen reference: `ACD-v5`.
- Frozen decisive split: `combined_abort_stress`.
- Frozen aggregate: hidden-wall, force-limit, human-stop, and combined stress.
- Frozen risk budgets: [0.08, 0.12, 0.18, 0.25].
- Bright boxed citation links.
- Numbered PDF copied only to Downloads.

B ALL MAIN METRICS

Table 9: All split-by-method metrics

Method	Split	Success	Repeated	Violation	Boundary F1	Efficiency
oracle	combined	0.911 ± 0.044	0.205 ± 0.093	0.295 ± 0.093	1.000 ± 0.000	0.378 ± 0.024
barrier-MPC	combined	0.884 ± 0.052	0.250 ± 0.046	0.366 ± 0.107	0.396 ± 0.009	0.318 ± 0.024
particle	combined	0.866 ± 0.072	0.259 ± 0.091	0.321 ± 0.095	0.396 ± 0.010	0.318 ± 0.036
kernel-cls	combined	0.857 ± 0.059	0.357 ± 0.099	0.482 ± 0.121	0.322 ± 0.010	0.295 ± 0.037
ACD-v4	combined	0.732 ± 0.044	0.366 ± 0.081	0.536 ± 0.088	0.446 ± 0.012	0.241 ± 0.027
conformal	combined	0.661 ± 0.051	0.473 ± 0.106	0.455 ± 0.079	0.407 ± 0.010	0.284 ± 0.031
trace-cls	combined	0.607 ± 0.059	0.509 ± 0.104	0.500 ± 0.102	0.393 ± 0.010	0.273 ± 0.031
ACD-v5	combined	0.545 ± 0.059	0.170 ± 0.079	0.259 ± 0.083	0.432 ± 0.012	0.183 ± 0.023
costmap	combined	0.536 ± 0.070	0.705 ± 0.067	0.777 ± 0.072	0.424 ± 0.011	0.198 ± 0.022
risk-filter	combined	0.500 ± 0.075	0.670 ± 0.087	0.705 ± 0.089	0.416 ± 0.009	0.207 ± 0.035
endpoint-neg	combined	0.188 ± 0.064	0.866 ± 0.041	0.920 ± 0.041	0.247 ± 0.007	0.123 ± 0.019
ignore	combined	0.027 ± 0.026	0.982 ± 0.023	0.848 ± 0.089	0.000 ± 0.000	0.091 ± 0.010
ACD-v4	force	1.000 ± 0.000	0.000 ± 0.000	0.045 ± 0.045	0.426 ± 0.014	0.579 ± 0.018
ACD-v5	force	1.000 ± 0.000	0.000 ± 0.000	0.036 ± 0.037	0.386 ± 0.012	0.560 ± 0.017
conformal	force	1.000 ± 0.000	0.009 ± 0.018	0.062 ± 0.032	0.347 ± 0.018	0.660 ± 0.013
trace-cls	force	1.000 ± 0.000	0.009 ± 0.018	0.089 ± 0.051	0.354 ± 0.016	0.648 ± 0.025
costmap	force	1.000 ± 0.000	0.000 ± 0.000	0.054 ± 0.044	0.383 ± 0.018	0.667 ± 0.013
kernel-cls	force	1.000 ± 0.000	0.009 ± 0.018	0.098 ± 0.052	0.305 ± 0.013	0.640 ± 0.019
particle	force	1.000 ± 0.000	0.000 ± 0.000	0.045 ± 0.045	0.369 ± 0.014	0.617 ± 0.015
risk-filter	force	1.000 ± 0.000	0.009 ± 0.018	0.071 ± 0.037	0.366 ± 0.015	0.660 ± 0.007
barrier-MPC	force	1.000 ± 0.000	0.000 ± 0.000	0.045 ± 0.037	0.345 ± 0.014	0.596 ± 0.023
endpoint-neg	force	0.973 ± 0.037	0.116 ± 0.052	0.179 ± 0.070	0.184 ± 0.015	0.638 ± 0.030
oracle	force	0.955 ± 0.037	0.125 ± 0.074	0.179 ± 0.092	1.000 ± 0.000	0.628 ± 0.033
ignore	force	0.714 ± 0.070	0.321 ± 0.070	0.375 ± 0.058	0.000 ± 0.000	0.517 ± 0.036
kernel-cls	hidden-wall	0.991 ± 0.018	0.134 ± 0.049	0.214 ± 0.059	0.257 ± 0.006	0.567 ± 0.025
oracle	hidden-wall	0.982 ± 0.035	0.062 ± 0.041	0.071 ± 0.046	1.000 ± 0.000	0.663 ± 0.023
particle	hidden-wall	0.982 ± 0.023	0.134 ± 0.056	0.196 ± 0.091	0.353 ± 0.010	0.540 ± 0.037
risk-filter	hidden-wall	0.929 ± 0.053	0.268 ± 0.087	0.304 ± 0.069	0.432 ± 0.009	0.532 ± 0.035
barrier-MPC	hidden-wall	0.929 ± 0.037	0.170 ± 0.070	0.304 ± 0.098	0.348 ± 0.009	0.519 ± 0.036
conformal	hidden-wall	0.911 ± 0.023	0.179 ± 0.046	0.250 ± 0.059	0.430 ± 0.010	0.565 ± 0.032
costmap	hidden-wall	0.911 ± 0.051	0.357 ± 0.059	0.420 ± 0.072	0.469 ± 0.007	0.496 ± 0.039
trace-cls	hidden-wall	0.893 ± 0.037	0.170 ± 0.052	0.259 ± 0.095	0.310 ± 0.007	0.561 ± 0.037
ACD-v4	hidden-wall	0.884 ± 0.052	0.223 ± 0.056	0.366 ± 0.056	0.345 ± 0.009	0.446 ± 0.024
ACD-v5	hidden-wall	0.723 ± 0.104	0.089 ± 0.058	0.125 ± 0.069	0.319 ± 0.012	0.375 ± 0.060

Table 9 continued

Method	Split	Success	Repeated	Violation	Boundary F1	Efficiency
endpoint-neg	hidden-wall	0.464 ± 0.079	0.670 ± 0.083	0.714 ± 0.065	0.374 ± 0.011	0.315 ± 0.044
ignore	hidden-wall	0.134 ± 0.049	0.893 ± 0.046	0.902 ± 0.052	0.000 ± 0.000	0.174 ± 0.034
ACD-v4	human-stop	1.000 ± 0.000	0.045 ± 0.045	0.170 ± 0.059	0.489 ± 0.021	0.517 ± 0.017
ACD-v5	human-stop	1.000 ± 0.000	0.045 ± 0.026	0.143 ± 0.065	0.485 ± 0.022	0.510 ± 0.018
conformal	human-stop	1.000 ± 0.000	0.036 ± 0.037	0.152 ± 0.041	0.326 ± 0.017	0.545 ± 0.023
trace-cls	human-stop	1.000 ± 0.000	0.062 ± 0.056	0.134 ± 0.032	0.354 ± 0.015	0.544 ± 0.025
costmap	human-stop	1.000 ± 0.000	0.089 ± 0.069	0.125 ± 0.058	0.382 ± 0.019	0.554 ± 0.020
kernel-cls	human-stop	1.000 ± 0.000	0.018 ± 0.023	0.080 ± 0.032	0.309 ± 0.012	0.562 ± 0.022
particle	human-stop	1.000 ± 0.000	0.036 ± 0.037	0.098 ± 0.037	0.370 ± 0.013	0.521 ± 0.027
risk-filter	human-stop	1.000 ± 0.000	0.089 ± 0.063	0.027 ± 0.026	0.368 ± 0.020	0.559 ± 0.023
barrier-MPC	human-stop	1.000 ± 0.000	0.027 ± 0.026	0.107 ± 0.046	0.360 ± 0.011	0.503 ± 0.023
oracle	human-stop	0.991 ± 0.018	0.062 ± 0.041	0.152 ± 0.049	1.000 ± 0.000	0.597 ± 0.012
endpoint-neg	human-stop	0.884 ± 0.064	0.214 ± 0.065	0.062 ± 0.032	0.166 ± 0.019	0.514 ± 0.040
ignore	human-stop	0.562 ± 0.056	0.482 ± 0.051	0.134 ± 0.081	0.000 ± 0.000	0.384 ± 0.030
ACD-v4	nominal	1.000 ± 0.000	0.000 ± 0.000	0.009 ± 0.018	0.777 ± 0.085	0.876 ± 0.007
ACD-v5	nominal	1.000 ± 0.000	0.000 ± 0.000	0.009 ± 0.018	0.777 ± 0.093	0.875 ± 0.008
conformal	nominal	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.786 ± 0.088	0.869 ± 0.009
trace-cls	nominal	1.000 ± 0.000	0.000 ± 0.000	0.018 ± 0.023	0.768 ± 0.091	0.868 ± 0.013
costmap	nominal	1.000 ± 0.000	0.000 ± 0.000	0.009 ± 0.018	0.786 ± 0.088	0.874 ± 0.011
ignore	nominal	1.000 ± 0.000	0.000 ± 0.000	0.009 ± 0.018	1.000 ± 0.000	0.877 ± 0.009
kernel-cls	nominal	1.000 ± 0.000	0.000 ± 0.000	0.018 ± 0.023	0.777 ± 0.093	0.862 ± 0.014
endpoint-neg	nominal	1.000 ± 0.000	0.009 ± 0.018	0.018 ± 0.023	0.768 ± 0.091	0.865 ± 0.010
oracle	nominal	1.000 ± 0.000	0.018 ± 0.023	0.054 ± 0.044	1.000 ± 0.000	0.860 ± 0.015
particle	nominal	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.786 ± 0.088	0.862 ± 0.012
risk-filter	nominal	1.000 ± 0.000	0.000 ± 0.000	0.009 ± 0.018	0.777 ± 0.093	0.868 ± 0.010
barrier-MPC	nominal	1.000 ± 0.000	0.000 ± 0.000	0.009 ± 0.018	0.777 ± 0.085	0.856 ± 0.016

C ALL MAIN PAIRWISE COMPARISONS

Table 10: All paired comparisons versus ACD-v5

Split	Comparison	Succ diff	CI	Viol red	Repeat red	F1 diff	Wins
combined	ACD-v4	-0.188	0.059	0.277	0.196	-0.013	0/8
combined	conformal	-0.116	0.064	0.196	0.304	0.025	1/8
combined	trace-cls	-0.062	0.062	0.241	0.339	0.040	2/8
combined	costmap	0.009	0.089	0.518	0.536	0.008	3/8
combined	ignore	0.518	0.078	0.589	0.812	0.432	8/8
combined	kernel-cls	-0.312	0.064	0.223	0.188	0.111	0/8
combined	endpoint-neg	0.357	0.070	0.661	0.696	0.185	8/8
combined	oracle	-0.366	0.056	0.036	0.036	-0.568	0/8
combined	particle	-0.321	0.065	0.062	0.089	0.036	0/8
combined	risk-filter	0.045	0.075	0.446	0.500	0.016	5/8
combined	barrier-MPC	-0.339	0.074	0.107	0.080	0.036	0/8
force	ACD-v4	0.000	0.000	0.009	0.000	-0.040	0/8
force	conformal	0.000	0.000	0.027	0.009	0.039	0/8
force	trace-cls	0.000	0.000	0.054	0.009	0.032	0/8
force	costmap	0.000	0.000	0.018	0.000	0.003	0/8
force	ignore	0.286	0.070	0.339	0.321	0.386	8/8
force	kernel-cls	0.000	0.000	0.062	0.009	0.082	0/8
force	endpoint-neg	0.027	0.037	0.143	0.116	0.202	2/8
force	oracle	0.045	0.037	0.143	0.125	-0.614	4/8
force	particle	0.000	0.000	0.009	0.000	0.018	0/8
force	risk-filter	0.000	0.000	0.036	0.009	0.020	0/8
force	barrier-MPC	0.000	0.000	0.009	0.000	0.041	0/8
hidden-wall	ACD-v4	-0.161	0.078	0.241	0.134	-0.026	0/8
hidden-wall	conformal	-0.188	0.112	0.125	0.089	-0.112	0/8
hidden-wall	trace-cls	-0.170	0.083	0.134	0.080	0.008	0/8
hidden-wall	costmap	-0.188	0.102	0.295	0.268	-0.151	1/8
hidden-wall	ignore	0.589	0.102	0.777	0.804	0.319	8/8
hidden-wall	kernel-cls	-0.268	0.105	0.089	0.045	0.061	0/8
hidden-wall	endpoint-neg	0.259	0.140	0.589	0.580	-0.056	8/8
hidden-wall	oracle	-0.259	0.109	-0.054	-0.027	-0.681	0/8
hidden-wall	particle	-0.259	0.112	0.071	0.045	-0.034	0/8
hidden-wall	risk-filter	-0.205	0.128	0.179	0.179	-0.114	1/8
hidden-wall	barrier-MPC	-0.205	0.114	0.179	0.080	-0.029	0/8
human-stop	ACD-v4	0.000	0.000	0.027	0.000	-0.005	0/8

Table 10 continued

Split	Comparison	Succ diff	CI	Viol red	Repeat red	F1 diff	Wins
human-stop	conformal	0.000	0.000	0.009	-0.009	0.159	0/8
human-stop	trace-cls	0.000	0.000	-0.009	0.018	0.131	0/8
human-stop	costmap	0.000	0.000	-0.018	0.045	0.102	0/8
human-stop	ignore	0.438	0.056	-0.009	0.438	0.485	8/8
human-stop	kernel-cls	0.000	0.000	-0.062	-0.027	0.176	0/8
human-stop	endpoint-neg	0.116	0.064	-0.080	0.170	0.319	7/8
human-stop	oracle	0.009	0.018	0.009	0.018	-0.515	1/8
human-stop	particle	0.000	0.000	-0.045	-0.009	0.114	0/8
human-stop	risk-filter	0.000	0.000	-0.116	0.045	0.117	0/8
human-stop	barrier-MPC	0.000	0.000	-0.036	-0.018	0.125	0/8
nominal	ACD-v4	0.000	0.000	0.000	0.000	0.000	0/8
nominal	conformal	0.000	0.000	-0.009	0.000	-0.009	0/8
nominal	trace-cls	0.000	0.000	0.009	0.000	0.009	0/8
nominal	costmap	0.000	0.000	0.000	0.000	-0.009	0/8
nominal	ignore	0.000	0.000	0.000	0.000	-0.223	0/8
nominal	kernel-cls	0.000	0.000	0.009	0.000	0.000	0/8
nominal	endpoint-neg	0.000	0.000	0.009	0.009	0.009	0/8
nominal	oracle	0.000	0.000	0.045	0.018	-0.223	0/8
nominal	particle	0.000	0.000	-0.009	0.000	-0.009	0/8
nominal	risk-filter	0.000	0.000	0.000	0.000	0.000	0/8
nominal	barrier-MPC	0.000	0.000	0.000	0.000	0.000	0/8

D Seed-Level Main Metrics

Table 11: Seed-level main metrics

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
ACD-v4	combined	0	14	0.643	0.500	0.571	0.463	0.222
ACD-v4	combined	1	14	0.786	0.214	0.500	0.411	0.281
ACD-v4	combined	2	14	0.714	0.286	0.357	0.452	0.297
ACD-v4	combined	3	14	0.786	0.357	0.571	0.427	0.225
ACD-v4	combined	4	14	0.786	0.286	0.571	0.455	0.251
ACD-v4	combined	5	14	0.714	0.286	0.429	0.448	0.249
ACD-v4	combined	6	14	0.786	0.500	0.500	0.452	0.236
ACD-v4	combined	7	14	0.643	0.500	0.786	0.458	0.168
ACD-v5	combined	0	14	0.429	0.214	0.286	0.436	0.180
ACD-v5	combined	1	14	0.429	0.071	0.143	0.398	0.166
ACD-v5	combined	2	14	0.500	0.214	0.214	0.443	0.167
ACD-v5	combined	3	14	0.643	0.071	0.143	0.418	0.175
ACD-v5	combined	4	14	0.571	0.071	0.143	0.446	0.153
ACD-v5	combined	5	14	0.571	0.357	0.429	0.432	0.205
ACD-v5	combined	6	14	0.643	0.071	0.286	0.442	0.255
ACD-v5	combined	7	14	0.571	0.286	0.429	0.445	0.164
conformal	combined	0	14	0.571	0.714	0.571	0.415	0.218
conformal	combined	1	14	0.643	0.643	0.643	0.384	0.249
conformal	combined	2	14	0.643	0.500	0.500	0.406	0.271
conformal	combined	3	14	0.714	0.429	0.429	0.390	0.295
conformal	combined	4	14	0.714	0.286	0.286	0.418	0.350
conformal	combined	5	14	0.786	0.286	0.429	0.401	0.335
conformal	combined	6	14	0.571	0.500	0.357	0.424	0.259
conformal	combined	7	14	0.643	0.429	0.429	0.418	0.300
trace-cls	combined	0	14	0.500	0.714	0.571	0.398	0.213
trace-cls	combined	1	14	0.500	0.643	0.643	0.381	0.239
trace-cls	combined	2	14	0.643	0.643	0.643	0.386	0.243
trace-cls	combined	3	14	0.571	0.500	0.571	0.370	0.267
trace-cls	combined	4	14	0.714	0.286	0.286	0.404	0.359
trace-cls	combined	5	14	0.714	0.357	0.571	0.389	0.295
trace-cls	combined	6	14	0.571	0.429	0.286	0.414	0.290
trace-cls	combined	7	14	0.643	0.500	0.429	0.399	0.276
costmap	combined	0	14	0.500	0.786	0.857	0.439	0.177
costmap	combined	1	14	0.429	0.857	0.929	0.403	0.150
costmap	combined	2	14	0.643	0.714	0.857	0.419	0.201
costmap	combined	3	14	0.429	0.714	0.714	0.400	0.187
costmap	combined	4	14	0.571	0.571	0.643	0.435	0.251
costmap	combined	5	14	0.500	0.714	0.786	0.423	0.192

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
costmap	combined	6	14	0.500	0.714	0.643	0.438	0.191
costmap	combined	7	14	0.714	0.571	0.786	0.436	0.235
ignore	combined	0	14	0.071	0.929	0.643	0.000	0.115
ignore	combined	1	14	0.071	0.929	0.714	0.000	0.111
ignore	combined	2	14	0.000	1.000	0.786	0.000	0.080
ignore	combined	3	14	0.000	1.000	1.000	0.000	0.092
ignore	combined	4	14	0.000	1.000	1.000	0.000	0.078
ignore	combined	5	14	0.000	1.000	0.929	0.000	0.083
ignore	combined	6	14	0.000	1.000	0.857	0.000	0.085
ignore	combined	7	14	0.071	1.000	0.857	0.000	0.083
kernel-cls	combined	0	14	0.786	0.429	0.643	0.323	0.250
kernel-cls	combined	1	14	0.714	0.571	0.714	0.316	0.225
kernel-cls	combined	2	14	0.929	0.357	0.429	0.310	0.305
kernel-cls	combined	3	14	0.786	0.357	0.429	0.297	0.293
kernel-cls	combined	4	14	0.929	0.071	0.143	0.340	0.399
kernel-cls	combined	5	14	0.929	0.357	0.500	0.321	0.302
kernel-cls	combined	6	14	0.857	0.429	0.571	0.327	0.263
kernel-cls	combined	7	14	0.929	0.286	0.429	0.340	0.327
endpoint-neg	combined	0	14	0.143	0.857	0.929	0.248	0.112
endpoint-neg	combined	1	14	0.143	0.857	0.857	0.237	0.120
endpoint-neg	combined	2	14	0.143	0.929	0.857	0.250	0.110
endpoint-neg	combined	3	14	0.286	0.786	0.857	0.239	0.172
endpoint-neg	combined	4	14	0.071	0.929	1.000	0.257	0.089
endpoint-neg	combined	5	14	0.357	0.786	0.929	0.236	0.158
endpoint-neg	combined	6	14	0.143	0.929	0.929	0.266	0.113
endpoint-neg	combined	7	14	0.214	0.857	1.000	0.245	0.113
oracle	combined	0	14	0.929	0.143	0.286	1.000	0.381
oracle	combined	1	14	0.786	0.500	0.571	1.000	0.306
oracle	combined	2	14	0.929	0.214	0.357	1.000	0.375
oracle	combined	3	14	0.929	0.071	0.143	1.000	0.429
oracle	combined	4	14	0.857	0.143	0.286	1.000	0.376
oracle	combined	5	14	1.000	0.143	0.143	1.000	0.403
oracle	combined	6	14	0.929	0.143	0.286	1.000	0.378
oracle	combined	7	14	0.929	0.286	0.286	1.000	0.377
particle	combined	0	14	0.786	0.500	0.571	0.398	0.241
particle	combined	1	14	0.714	0.286	0.357	0.382	0.272
particle	combined	2	14	0.857	0.214	0.214	0.386	0.325
particle	combined	3	14	0.929	0.286	0.357	0.375	0.321

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
particle	combined	4	14	1.000	0.071	0.143	0.417	0.409
particle	combined	5	14	1.000	0.143	0.286	0.395	0.326
particle	combined	6	14	0.786	0.357	0.429	0.412	0.287
particle	combined	7	14	0.857	0.214	0.214	0.405	0.358
risk-filter	combined	0	14	0.357	0.857	0.857	0.431	0.129
risk-filter	combined	1	14	0.500	0.714	0.786	0.399	0.212
risk-filter	combined	2	14	0.429	0.714	0.643	0.412	0.178
risk-filter	combined	3	14	0.429	0.714	0.786	0.396	0.196
risk-filter	combined	4	14	0.643	0.429	0.571	0.425	0.304
risk-filter	combined	5	14	0.429	0.714	0.857	0.413	0.182
risk-filter	combined	6	14	0.571	0.571	0.571	0.423	0.228
risk-filter	combined	7	14	0.643	0.643	0.571	0.431	0.229
barrier-MPC	combined	0	14	0.786	0.357	0.500	0.399	0.270
barrier-MPC	combined	1	14	0.929	0.214	0.643	0.386	0.275
barrier-MPC	combined	2	14	0.929	0.214	0.286	0.380	0.349
barrier-MPC	combined	3	14	0.929	0.214	0.286	0.379	0.329
barrier-MPC	combined	4	14	1.000	0.143	0.143	0.411	0.372
barrier-MPC	combined	5	14	0.857	0.286	0.286	0.395	0.304
barrier-MPC	combined	6	14	0.857	0.286	0.357	0.412	0.318
barrier-MPC	combined	7	14	0.786	0.286	0.429	0.407	0.323
ACD-v4	force	0	14	1.000	0.000	0.000	0.417	0.579
ACD-v4	force	1	14	1.000	0.000	0.000	0.420	0.577
ACD-v4	force	2	14	1.000	0.000	0.000	0.392	0.599
ACD-v4	force	3	14	1.000	0.000	0.143	0.445	0.542
ACD-v4	force	4	14	1.000	0.000	0.071	0.456	0.615
ACD-v4	force	5	14	1.000	0.000	0.000	0.427	0.565
ACD-v4	force	6	14	1.000	0.000	0.000	0.413	0.551
ACD-v4	force	7	14	1.000	0.000	0.143	0.438	0.604
ACD-v5	force	0	14	1.000	0.000	0.071	0.378	0.540
ACD-v5	force	1	14	1.000	0.000	0.071	0.381	0.562
ACD-v5	force	2	14	1.000	0.000	0.000	0.359	0.597
ACD-v5	force	3	14	1.000	0.000	0.000	0.410	0.532
ACD-v5	force	4	14	1.000	0.000	0.000	0.407	0.590
ACD-v5	force	5	14	1.000	0.000	0.000	0.395	0.544
ACD-v5	force	6	14	1.000	0.000	0.000	0.370	0.540
ACD-v5	force	7	14	1.000	0.000	0.143	0.390	0.576
conformal	force	0	14	1.000	0.000	0.071	0.364	0.647
conformal	force	1	14	1.000	0.000	0.071	0.326	0.648

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
conformal	force	2	14	1.000	0.000	0.000	0.331	0.685
conformal	force	3	14	1.000	0.000	0.071	0.315	0.663
conformal	force	4	14	1.000	0.000	0.143	0.337	0.654
conformal	force	5	14	1.000	0.071	0.071	0.353	0.629
conformal	force	6	14	1.000	0.000	0.071	0.397	0.671
conformal	force	7	14	1.000	0.000	0.000	0.352	0.683
trace-cls	force	0	14	1.000	0.000	0.000	0.367	0.652
trace-cls	force	1	14	1.000	0.000	0.214	0.324	0.593
trace-cls	force	2	14	1.000	0.000	0.071	0.343	0.663
trace-cls	force	3	14	1.000	0.000	0.071	0.323	0.679
trace-cls	force	4	14	1.000	0.000	0.143	0.357	0.627
trace-cls	force	5	14	1.000	0.071	0.143	0.383	0.620
trace-cls	force	6	14	1.000	0.000	0.071	0.354	0.644
trace-cls	force	7	14	1.000	0.000	0.000	0.384	0.706
costmap	force	0	14	1.000	0.000	0.000	0.399	0.666
costmap	force	1	14	1.000	0.000	0.143	0.362	0.642
costmap	force	2	14	1.000	0.000	0.143	0.352	0.646
costmap	force	3	14	1.000	0.000	0.071	0.373	0.662
costmap	force	4	14	1.000	0.000	0.000	0.385	0.699
costmap	force	5	14	1.000	0.000	0.071	0.371	0.662
costmap	force	6	14	1.000	0.000	0.000	0.435	0.676
costmap	force	7	14	1.000	0.000	0.000	0.385	0.685
ignore	force	0	14	0.643	0.500	0.500	0.000	0.468
ignore	force	1	14	0.643	0.357	0.429	0.000	0.472
ignore	force	2	14	0.714	0.286	0.286	0.000	0.531
ignore	force	3	14	0.714	0.286	0.429	0.000	0.526
ignore	force	4	14	0.857	0.214	0.357	0.000	0.571
ignore	force	5	14	0.571	0.429	0.429	0.000	0.444
ignore	force	6	14	0.714	0.286	0.286	0.000	0.541
ignore	force	7	14	0.857	0.214	0.286	0.000	0.588
kernel-cls	force	0	14	1.000	0.000	0.000	0.311	0.669
kernel-cls	force	1	14	1.000	0.000	0.143	0.300	0.618
kernel-cls	force	2	14	1.000	0.000	0.143	0.274	0.637
kernel-cls	force	3	14	1.000	0.000	0.214	0.291	0.595
kernel-cls	force	4	14	1.000	0.071	0.071	0.288	0.661
kernel-cls	force	5	14	1.000	0.000	0.143	0.326	0.616
kernel-cls	force	6	14	1.000	0.000	0.000	0.324	0.658
kernel-cls	force	7	14	1.000	0.000	0.071	0.323	0.665

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
endpoint-neg	force	0	14	0.857	0.143	0.357	0.213	0.574
endpoint-neg	force	1	14	1.000	0.214	0.214	0.203	0.602
endpoint-neg	force	2	14	0.929	0.143	0.143	0.177	0.622
endpoint-neg	force	3	14	1.000	0.071	0.143	0.197	0.646
endpoint-neg	force	4	14	1.000	0.071	0.143	0.151	0.659
endpoint-neg	force	5	14	1.000	0.214	0.214	0.181	0.635
endpoint-neg	force	6	14	1.000	0.000	0.000	0.195	0.719
endpoint-neg	force	7	14	1.000	0.071	0.214	0.157	0.644
oracle	force	0	14	0.929	0.214	0.286	1.000	0.591
oracle	force	1	14	1.000	0.071	0.143	1.000	0.651
oracle	force	2	14	1.000	0.071	0.143	1.000	0.660
oracle	force	3	14	0.929	0.143	0.214	1.000	0.623
oracle	force	4	14	0.929	0.214	0.286	1.000	0.586
oracle	force	5	14	1.000	0.000	0.000	1.000	0.689
oracle	force	6	14	1.000	0.000	0.000	1.000	0.671
oracle	force	7	14	0.857	0.286	0.357	1.000	0.554
particle	force	0	14	1.000	0.000	0.071	0.381	0.612
particle	force	1	14	1.000	0.000	0.000	0.341	0.611
particle	force	2	14	1.000	0.000	0.143	0.343	0.589
particle	force	3	14	1.000	0.000	0.143	0.354	0.593
particle	force	4	14	1.000	0.000	0.000	0.372	0.636
particle	force	5	14	1.000	0.000	0.000	0.381	0.639
particle	force	6	14	1.000	0.000	0.000	0.388	0.607
particle	force	7	14	1.000	0.000	0.000	0.388	0.649
risk-filter	force	0	14	1.000	0.000	0.000	0.383	0.669
risk-filter	force	1	14	1.000	0.000	0.143	0.357	0.638
risk-filter	force	2	14	1.000	0.000	0.071	0.337	0.664
risk-filter	force	3	14	1.000	0.000	0.000	0.357	0.669
risk-filter	force	4	14	1.000	0.000	0.071	0.369	0.657
risk-filter	force	5	14	1.000	0.000	0.143	0.354	0.657
risk-filter	force	6	14	1.000	0.071	0.071	0.408	0.656
risk-filter	force	7	14	1.000	0.000	0.071	0.361	0.670
barrier-MPC	force	0	14	1.000	0.000	0.143	0.345	0.534
barrier-MPC	force	1	14	1.000	0.000	0.071	0.314	0.591
barrier-MPC	force	2	14	1.000	0.000	0.000	0.328	0.639
barrier-MPC	force	3	14	1.000	0.000	0.071	0.327	0.578
barrier-MPC	force	4	14	1.000	0.000	0.000	0.364	0.629
barrier-MPC	force	5	14	1.000	0.000	0.000	0.357	0.607

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
barrier-MPC	force	6	14	1.000	0.000	0.071	0.363	0.580
barrier-MPC	force	7	14	1.000	0.000	0.000	0.365	0.611
ACD-v4	hidden-wall	0	14	0.786	0.286	0.357	0.331	0.404
ACD-v4	hidden-wall	1	14	1.000	0.214	0.286	0.336	0.456
ACD-v4	hidden-wall	2	14	0.857	0.286	0.429	0.346	0.415
ACD-v4	hidden-wall	3	14	0.857	0.071	0.429	0.351	0.441
ACD-v4	hidden-wall	4	14	0.786	0.286	0.357	0.336	0.417
ACD-v4	hidden-wall	5	14	0.929	0.286	0.286	0.369	0.470
ACD-v4	hidden-wall	6	14	0.929	0.143	0.286	0.334	0.506
ACD-v4	hidden-wall	7	14	0.929	0.214	0.500	0.356	0.462
ACD-v5	hidden-wall	0	14	0.714	0.214	0.143	0.303	0.331
ACD-v5	hidden-wall	1	14	0.857	0.143	0.214	0.306	0.438
ACD-v5	hidden-wall	2	14	0.500	0.143	0.143	0.324	0.265
ACD-v5	hidden-wall	3	14	0.643	0.000	0.000	0.326	0.347
ACD-v5	hidden-wall	4	14	0.571	0.071	0.143	0.308	0.269
ACD-v5	hidden-wall	5	14	0.714	0.000	0.000	0.354	0.390
ACD-v5	hidden-wall	6	14	0.857	0.000	0.071	0.302	0.495
ACD-v5	hidden-wall	7	14	0.929	0.143	0.286	0.326	0.462
conformal	hidden-wall	0	14	0.929	0.071	0.286	0.423	0.591
conformal	hidden-wall	1	14	0.857	0.214	0.286	0.404	0.483
conformal	hidden-wall	2	14	0.929	0.214	0.214	0.435	0.553
conformal	hidden-wall	3	14	0.929	0.214	0.357	0.442	0.551
conformal	hidden-wall	4	14	0.929	0.143	0.143	0.426	0.625
conformal	hidden-wall	5	14	0.857	0.286	0.357	0.446	0.527
conformal	hidden-wall	6	14	0.929	0.143	0.214	0.422	0.588
conformal	hidden-wall	7	14	0.929	0.143	0.143	0.444	0.604
trace-cls	hidden-wall	0	14	0.929	0.071	0.286	0.307	0.598
trace-cls	hidden-wall	1	14	0.857	0.214	0.429	0.296	0.479
trace-cls	hidden-wall	2	14	0.857	0.214	0.214	0.305	0.553
trace-cls	hidden-wall	3	14	0.857	0.143	0.357	0.305	0.528
trace-cls	hidden-wall	4	14	0.857	0.143	0.143	0.315	0.619
trace-cls	hidden-wall	5	14	0.857	0.214	0.286	0.320	0.537
trace-cls	hidden-wall	6	14	0.929	0.286	0.357	0.305	0.537
trace-cls	hidden-wall	7	14	1.000	0.071	0.000	0.329	0.640
costmap	hidden-wall	0	14	0.857	0.429	0.500	0.463	0.458
costmap	hidden-wall	1	14	0.929	0.500	0.571	0.450	0.434
costmap	hidden-wall	2	14	0.786	0.357	0.429	0.474	0.451
costmap	hidden-wall	3	14	1.000	0.214	0.357	0.476	0.589

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
costmap	hidden-wall	4	14	0.929	0.286	0.286	0.475	0.525
costmap	hidden-wall	5	14	0.929	0.357	0.500	0.463	0.484
costmap	hidden-wall	6	14	1.000	0.357	0.286	0.485	0.560
costmap	hidden-wall	7	14	0.857	0.357	0.429	0.470	0.463
ignore	hidden-wall	0	14	0.214	0.857	0.857	0.000	0.193
ignore	hidden-wall	1	14	0.071	0.929	1.000	0.000	0.124
ignore	hidden-wall	2	14	0.071	1.000	1.000	0.000	0.117
ignore	hidden-wall	3	14	0.214	0.857	0.857	0.000	0.227
ignore	hidden-wall	4	14	0.071	0.929	0.929	0.000	0.145
ignore	hidden-wall	5	14	0.071	0.929	0.929	0.000	0.148
ignore	hidden-wall	6	14	0.214	0.786	0.786	0.000	0.249
ignore	hidden-wall	7	14	0.143	0.857	0.857	0.000	0.194
kernel-cls	hidden-wall	0	14	0.929	0.214	0.286	0.248	0.537
kernel-cls	hidden-wall	1	14	1.000	0.143	0.357	0.246	0.504
kernel-cls	hidden-wall	2	14	1.000	0.000	0.214	0.265	0.578
kernel-cls	hidden-wall	3	14	1.000	0.143	0.214	0.252	0.571
kernel-cls	hidden-wall	4	14	1.000	0.143	0.214	0.258	0.614
kernel-cls	hidden-wall	5	14	1.000	0.071	0.071	0.261	0.564
kernel-cls	hidden-wall	6	14	1.000	0.214	0.214	0.262	0.561
kernel-cls	hidden-wall	7	14	1.000	0.143	0.143	0.268	0.608
endpoint-neg	hidden-wall	0	14	0.571	0.643	0.714	0.367	0.332
endpoint-neg	hidden-wall	1	14	0.357	0.714	0.714	0.371	0.274
endpoint-neg	hidden-wall	2	14	0.429	0.714	0.714	0.402	0.321
endpoint-neg	hidden-wall	3	14	0.500	0.643	0.786	0.381	0.331
endpoint-neg	hidden-wall	4	14	0.429	0.643	0.714	0.358	0.312
endpoint-neg	hidden-wall	5	14	0.500	0.714	0.786	0.354	0.288
endpoint-neg	hidden-wall	6	14	0.643	0.429	0.500	0.390	0.441
endpoint-neg	hidden-wall	7	14	0.286	0.857	0.786	0.373	0.221
oracle	hidden-wall	0	14	0.857	0.143	0.143	1.000	0.616
oracle	hidden-wall	1	14	1.000	0.000	0.000	1.000	0.695
oracle	hidden-wall	2	14	1.000	0.071	0.143	1.000	0.633
oracle	hidden-wall	3	14	1.000	0.143	0.143	1.000	0.648
oracle	hidden-wall	4	14	1.000	0.000	0.000	1.000	0.709
oracle	hidden-wall	5	14	1.000	0.000	0.000	1.000	0.696
oracle	hidden-wall	6	14	1.000	0.071	0.071	1.000	0.657
oracle	hidden-wall	7	14	1.000	0.071	0.071	1.000	0.649
particle	hidden-wall	0	14	0.929	0.286	0.357	0.336	0.494
particle	hidden-wall	1	14	0.929	0.214	0.357	0.326	0.445

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
particle	hidden-wall	2	14	1.000	0.071	0.286	0.356	0.552
particle	hidden-wall	3	14	1.000	0.143	0.143	0.355	0.538
particle	hidden-wall	4	14	1.000	0.071	0.071	0.353	0.584
particle	hidden-wall	5	14	1.000	0.071	0.000	0.367	0.572
particle	hidden-wall	6	14	1.000	0.143	0.214	0.357	0.519
particle	hidden-wall	7	14	1.000	0.071	0.143	0.371	0.612
risk-filter	hidden-wall	0	14	0.786	0.357	0.286	0.417	0.500
risk-filter	hidden-wall	1	14	0.929	0.429	0.357	0.413	0.465
risk-filter	hidden-wall	2	14	1.000	0.214	0.286	0.451	0.540
risk-filter	hidden-wall	3	14	0.929	0.143	0.214	0.436	0.584
risk-filter	hidden-wall	4	14	0.929	0.357	0.500	0.434	0.512
risk-filter	hidden-wall	5	14	1.000	0.071	0.214	0.424	0.614
risk-filter	hidden-wall	6	14	1.000	0.214	0.214	0.445	0.549
risk-filter	hidden-wall	7	14	0.857	0.357	0.357	0.436	0.491
barrier-MPC	hidden-wall	0	14	0.929	0.286	0.429	0.342	0.508
barrier-MPC	hidden-wall	1	14	0.857	0.214	0.429	0.318	0.452
barrier-MPC	hidden-wall	2	14	0.857	0.214	0.286	0.349	0.506
barrier-MPC	hidden-wall	3	14	0.929	0.286	0.500	0.348	0.445
barrier-MPC	hidden-wall	4	14	1.000	0.143	0.286	0.351	0.565
barrier-MPC	hidden-wall	5	14	1.000	0.000	0.071	0.358	0.596
barrier-MPC	hidden-wall	6	14	0.929	0.143	0.214	0.356	0.538
barrier-MPC	hidden-wall	7	14	0.929	0.071	0.214	0.362	0.546
ACD-v4	human-stop	0	14	1.000	0.000	0.214	0.475	0.525
ACD-v4	human-stop	1	14	1.000	0.000	0.143	0.480	0.563
ACD-v4	human-stop	2	14	1.000	0.000	0.071	0.500	0.497
ACD-v4	human-stop	3	14	1.000	0.000	0.143	0.478	0.518
ACD-v4	human-stop	4	14	1.000	0.143	0.143	0.506	0.494
ACD-v4	human-stop	5	14	1.000	0.000	0.286	0.435	0.515
ACD-v4	human-stop	6	14	1.000	0.143	0.286	0.537	0.488
ACD-v4	human-stop	7	14	1.000	0.071	0.071	0.505	0.534
ACD-v5	human-stop	0	14	1.000	0.071	0.214	0.466	0.528
ACD-v5	human-stop	1	14	1.000	0.000	0.000	0.480	0.565
ACD-v5	human-stop	2	14	1.000	0.000	0.071	0.488	0.485
ACD-v5	human-stop	3	14	1.000	0.000	0.143	0.488	0.504
ACD-v5	human-stop	4	14	1.000	0.071	0.214	0.510	0.492
ACD-v5	human-stop	5	14	1.000	0.071	0.286	0.421	0.493
ACD-v5	human-stop	6	14	1.000	0.071	0.143	0.521	0.514
ACD-v5	human-stop	7	14	1.000	0.071	0.071	0.506	0.497

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
conformal	human-stop	0	14	1.000	0.000	0.071	0.326	0.590
conformal	human-stop	1	14	1.000	0.000	0.143	0.318	0.581
conformal	human-stop	2	14	1.000	0.000	0.143	0.354	0.515
conformal	human-stop	3	14	1.000	0.143	0.214	0.296	0.495
conformal	human-stop	4	14	1.000	0.000	0.071	0.322	0.561
conformal	human-stop	5	14	1.000	0.071	0.214	0.288	0.534
conformal	human-stop	6	14	1.000	0.071	0.214	0.342	0.527
conformal	human-stop	7	14	1.000	0.000	0.143	0.357	0.559
trace-cls	human-stop	0	14	1.000	0.000	0.143	0.365	0.558
trace-cls	human-stop	1	14	1.000	0.071	0.071	0.342	0.609
trace-cls	human-stop	2	14	1.000	0.000	0.143	0.346	0.532
trace-cls	human-stop	3	14	1.000	0.143	0.214	0.342	0.493
trace-cls	human-stop	4	14	1.000	0.214	0.143	0.344	0.515
trace-cls	human-stop	5	14	1.000	0.071	0.143	0.322	0.552
trace-cls	human-stop	6	14	1.000	0.000	0.143	0.386	0.525
trace-cls	human-stop	7	14	1.000	0.000	0.071	0.383	0.568
costmap	human-stop	0	14	1.000	0.143	0.214	0.390	0.549
costmap	human-stop	1	14	1.000	0.071	0.214	0.373	0.548
costmap	human-stop	2	14	1.000	0.143	0.071	0.415	0.538
costmap	human-stop	3	14	1.000	0.286	0.214	0.378	0.501
costmap	human-stop	4	14	1.000	0.000	0.071	0.367	0.575
costmap	human-stop	5	14	1.000	0.071	0.143	0.329	0.572
costmap	human-stop	6	14	1.000	0.000	0.071	0.411	0.550
costmap	human-stop	7	14	1.000	0.000	0.000	0.397	0.601
ignore	human-stop	0	14	0.643	0.500	0.214	0.000	0.404
ignore	human-stop	1	14	0.643	0.357	0.000	0.000	0.446
ignore	human-stop	2	14	0.500	0.500	0.000	0.000	0.365
ignore	human-stop	3	14	0.571	0.429	0.143	0.000	0.376
ignore	human-stop	4	14	0.643	0.429	0.143	0.000	0.434
ignore	human-stop	5	14	0.571	0.500	0.143	0.000	0.387
ignore	human-stop	6	14	0.500	0.571	0.357	0.000	0.337
ignore	human-stop	7	14	0.429	0.571	0.071	0.000	0.324
kernel-cls	human-stop	0	14	1.000	0.000	0.071	0.291	0.620
kernel-cls	human-stop	1	14	1.000	0.000	0.071	0.307	0.580
kernel-cls	human-stop	2	14	1.000	0.000	0.071	0.314	0.540
kernel-cls	human-stop	3	14	1.000	0.000	0.143	0.290	0.525
kernel-cls	human-stop	4	14	1.000	0.071	0.143	0.296	0.575
kernel-cls	human-stop	5	14	1.000	0.071	0.071	0.309	0.537

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
kernel-cls	human-stop	6	14	1.000	0.000	0.071	0.330	0.542
kernel-cls	human-stop	7	14	1.000	0.000	0.000	0.335	0.578
endpoint-neg	human-stop	0	14	0.929	0.071	0.000	0.154	0.595
endpoint-neg	human-stop	1	14	0.929	0.286	0.071	0.181	0.480
endpoint-neg	human-stop	2	14	0.786	0.286	0.071	0.205	0.489
endpoint-neg	human-stop	3	14	0.714	0.357	0.143	0.181	0.422
endpoint-neg	human-stop	4	14	0.857	0.214	0.071	0.141	0.519
endpoint-neg	human-stop	5	14	1.000	0.143	0.071	0.119	0.594
endpoint-neg	human-stop	6	14	0.929	0.143	0.071	0.168	0.512
endpoint-neg	human-stop	7	14	0.929	0.214	0.000	0.180	0.498
oracle	human-stop	0	14	0.929	0.071	0.143	1.000	0.593
oracle	human-stop	1	14	1.000	0.000	0.214	1.000	0.599
oracle	human-stop	2	14	1.000	0.000	0.214	1.000	0.589
oracle	human-stop	3	14	1.000	0.143	0.214	1.000	0.576
oracle	human-stop	4	14	1.000	0.071	0.143	1.000	0.588
oracle	human-stop	5	14	1.000	0.071	0.143	1.000	0.608
oracle	human-stop	6	14	1.000	0.143	0.143	1.000	0.591
oracle	human-stop	7	14	1.000	0.000	0.000	1.000	0.635
particle	human-stop	0	14	1.000	0.000	0.071	0.363	0.559
particle	human-stop	1	14	1.000	0.000	0.000	0.366	0.574
particle	human-stop	2	14	1.000	0.071	0.143	0.376	0.487
particle	human-stop	3	14	1.000	0.143	0.143	0.379	0.453
particle	human-stop	4	14	1.000	0.000	0.071	0.355	0.531
particle	human-stop	5	14	1.000	0.071	0.143	0.340	0.517
particle	human-stop	6	14	1.000	0.000	0.071	0.395	0.521
particle	human-stop	7	14	1.000	0.000	0.143	0.389	0.529
risk-filter	human-stop	0	14	1.000	0.071	0.000	0.363	0.605
risk-filter	human-stop	1	14	1.000	0.071	0.000	0.382	0.547
risk-filter	human-stop	2	14	1.000	0.143	0.071	0.397	0.533
risk-filter	human-stop	3	14	1.000	0.286	0.000	0.369	0.501
risk-filter	human-stop	4	14	1.000	0.000	0.071	0.333	0.583
risk-filter	human-stop	5	14	1.000	0.071	0.071	0.319	0.545
risk-filter	human-stop	6	14	1.000	0.000	0.000	0.388	0.569
risk-filter	human-stop	7	14	1.000	0.071	0.000	0.392	0.586
barrier-MPC	human-stop	0	14	1.000	0.000	0.000	0.364	0.554
barrier-MPC	human-stop	1	14	1.000	0.000	0.143	0.340	0.505
barrier-MPC	human-stop	2	14	1.000	0.071	0.071	0.359	0.507
barrier-MPC	human-stop	3	14	1.000	0.071	0.143	0.361	0.435

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
barrier-MPC	human-stop	4	14	1.000	0.000	0.143	0.339	0.495
barrier-MPC	human-stop	5	14	1.000	0.000	0.214	0.360	0.505
barrier-MPC	human-stop	6	14	1.000	0.000	0.071	0.390	0.506
barrier-MPC	human-stop	7	14	1.000	0.071	0.071	0.368	0.515
ACD-v4	nominal	0	14	1.000	0.000	0.000	0.714	0.875
ACD-v4	nominal	1	14	1.000	0.000	0.000	0.857	0.888
ACD-v4	nominal	2	14	1.000	0.000	0.000	0.929	0.892
ACD-v4	nominal	3	14	1.000	0.000	0.000	0.714	0.874
ACD-v4	nominal	4	14	1.000	0.000	0.071	0.786	0.862
ACD-v4	nominal	5	14	1.000	0.000	0.000	0.571	0.870
ACD-v4	nominal	6	14	1.000	0.000	0.000	0.929	0.876
ACD-v4	nominal	7	14	1.000	0.000	0.000	0.714	0.870
ACD-v5	nominal	0	14	1.000	0.000	0.000	0.714	0.876
ACD-v5	nominal	1	14	1.000	0.000	0.000	0.857	0.883
ACD-v5	nominal	2	14	1.000	0.000	0.000	0.929	0.894
ACD-v5	nominal	3	14	1.000	0.000	0.071	0.643	0.859
ACD-v5	nominal	4	14	1.000	0.000	0.000	0.857	0.863
ACD-v5	nominal	5	14	1.000	0.000	0.000	0.571	0.872
ACD-v5	nominal	6	14	1.000	0.000	0.000	0.929	0.882
ACD-v5	nominal	7	14	1.000	0.000	0.000	0.714	0.866
conformal	nominal	0	14	1.000	0.000	0.000	0.714	0.846
conformal	nominal	1	14	1.000	0.000	0.000	0.857	0.875
conformal	nominal	2	14	1.000	0.000	0.000	0.929	0.884
conformal	nominal	3	14	1.000	0.000	0.000	0.714	0.882
conformal	nominal	4	14	1.000	0.000	0.000	0.857	0.865
conformal	nominal	5	14	1.000	0.000	0.000	0.571	0.869
conformal	nominal	6	14	1.000	0.000	0.000	0.929	0.875
conformal	nominal	7	14	1.000	0.000	0.000	0.714	0.855
trace-cls	nominal	0	14	1.000	0.000	0.000	0.714	0.854
trace-cls	nominal	1	14	1.000	0.000	0.000	0.857	0.890
trace-cls	nominal	2	14	1.000	0.000	0.000	0.929	0.889
trace-cls	nominal	3	14	1.000	0.000	0.071	0.643	0.840
trace-cls	nominal	4	14	1.000	0.000	0.071	0.786	0.850
trace-cls	nominal	5	14	1.000	0.000	0.000	0.571	0.873
trace-cls	nominal	6	14	1.000	0.000	0.000	0.929	0.880
trace-cls	nominal	7	14	1.000	0.000	0.000	0.714	0.864
costmap	nominal	0	14	1.000	0.000	0.000	0.714	0.875
costmap	nominal	1	14	1.000	0.000	0.000	0.857	0.891

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
costmap	nominal	2	14	1.000	0.000	0.000	0.929	0.895
costmap	nominal	3	14	1.000	0.000	0.000	0.714	0.875
costmap	nominal	4	14	1.000	0.000	0.000	0.857	0.867
costmap	nominal	5	14	1.000	0.000	0.071	0.571	0.849
costmap	nominal	6	14	1.000	0.000	0.000	0.929	0.879
costmap	nominal	7	14	1.000	0.000	0.000	0.714	0.860
ignore	nominal	0	14	1.000	0.000	0.000	1.000	0.878
ignore	nominal	1	14	1.000	0.000	0.000	1.000	0.896
ignore	nominal	2	14	1.000	0.000	0.000	1.000	0.887
ignore	nominal	3	14	1.000	0.000	0.000	1.000	0.886
ignore	nominal	4	14	1.000	0.000	0.071	1.000	0.863
ignore	nominal	5	14	1.000	0.000	0.000	1.000	0.862
ignore	nominal	6	14	1.000	0.000	0.000	1.000	0.876
ignore	nominal	7	14	1.000	0.000	0.000	1.000	0.865
kernel-cls	nominal	0	14	1.000	0.000	0.071	0.714	0.830
kernel-cls	nominal	1	14	1.000	0.000	0.000	0.857	0.882
kernel-cls	nominal	2	14	1.000	0.000	0.000	0.929	0.881
kernel-cls	nominal	3	14	1.000	0.000	0.071	0.643	0.842
kernel-cls	nominal	4	14	1.000	0.000	0.000	0.857	0.866
kernel-cls	nominal	5	14	1.000	0.000	0.000	0.571	0.868
kernel-cls	nominal	6	14	1.000	0.000	0.000	0.929	0.880
kernel-cls	nominal	7	14	1.000	0.000	0.000	0.714	0.851
endpoint-neg	nominal	0	14	1.000	0.000	0.000	0.714	0.867
endpoint-neg	nominal	1	14	1.000	0.000	0.000	0.857	0.879
endpoint-neg	nominal	2	14	1.000	0.000	0.000	0.929	0.882
endpoint-neg	nominal	3	14	1.000	0.071	0.071	0.643	0.842
endpoint-neg	nominal	4	14	1.000	0.000	0.071	0.786	0.855
endpoint-neg	nominal	5	14	1.000	0.000	0.000	0.571	0.867
endpoint-neg	nominal	6	14	1.000	0.000	0.000	0.929	0.877
endpoint-neg	nominal	7	14	1.000	0.000	0.000	0.714	0.855
oracle	nominal	0	14	1.000	0.000	0.071	1.000	0.861
oracle	nominal	1	14	1.000	0.000	0.071	1.000	0.858
oracle	nominal	2	14	1.000	0.000	0.000	1.000	0.885
oracle	nominal	3	14	1.000	0.071	0.143	1.000	0.823
oracle	nominal	4	14	1.000	0.000	0.000	1.000	0.875
oracle	nominal	5	14	1.000	0.000	0.000	1.000	0.862
oracle	nominal	6	14	1.000	0.000	0.000	1.000	0.879
oracle	nominal	7	14	1.000	0.071	0.143	1.000	0.837

Table 11 continued

Method	Split	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
particle	nominal	0	14	1.000	0.000	0.000	0.714	0.845
particle	nominal	1	14	1.000	0.000	0.000	0.857	0.873
particle	nominal	2	14	1.000	0.000	0.000	0.929	0.889
particle	nominal	3	14	1.000	0.000	0.000	0.714	0.870
particle	nominal	4	14	1.000	0.000	0.000	0.857	0.855
particle	nominal	5	14	1.000	0.000	0.000	0.571	0.847
particle	nominal	6	14	1.000	0.000	0.000	0.929	0.878
particle	nominal	7	14	1.000	0.000	0.000	0.714	0.842
risk-filter	nominal	0	14	1.000	0.000	0.000	0.714	0.864
risk-filter	nominal	1	14	1.000	0.000	0.000	0.857	0.879
risk-filter	nominal	2	14	1.000	0.000	0.000	0.929	0.887
risk-filter	nominal	3	14	1.000	0.000	0.071	0.643	0.844
risk-filter	nominal	4	14	1.000	0.000	0.000	0.857	0.864
risk-filter	nominal	5	14	1.000	0.000	0.000	0.571	0.866
risk-filter	nominal	6	14	1.000	0.000	0.000	0.929	0.878
risk-filter	nominal	7	14	1.000	0.000	0.000	0.714	0.858
barrier-MPC	nominal	0	14	1.000	0.000	0.000	0.714	0.844
barrier-MPC	nominal	1	14	1.000	0.000	0.000	0.857	0.873
barrier-MPC	nominal	2	14	1.000	0.000	0.000	0.929	0.888
barrier-MPC	nominal	3	14	1.000	0.000	0.000	0.714	0.854
barrier-MPC	nominal	4	14	1.000	0.000	0.071	0.786	0.835
barrier-MPC	nominal	5	14	1.000	0.000	0.000	0.571	0.849
barrier-MPC	nominal	6	14	1.000	0.000	0.000	0.929	0.885
barrier-MPC	nominal	7	14	1.000	0.000	0.000	0.714	0.822

E SEED-LEVEL AGGREGATE METRICS

Table 12: Seed-level aggregate metrics

Method	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
ACD-v4	0	56	0.857	0.196	0.286	0.422	0.432
ACD-v4	1	56	0.946	0.107	0.232	0.412	0.469
ACD-v4	2	56	0.893	0.143	0.214	0.422	0.452
ACD-v4	3	56	0.911	0.107	0.321	0.425	0.432
ACD-v4	4	56	0.893	0.179	0.286	0.438	0.444
ACD-v4	5	56	0.911	0.143	0.250	0.420	0.450
ACD-v4	6	56	0.929	0.196	0.268	0.434	0.445
ACD-v4	7	56	0.893	0.196	0.375	0.439	0.442
ACD-v5	0	56	0.786	0.125	0.179	0.395	0.395
ACD-v5	1	56	0.821	0.054	0.107	0.391	0.433
ACD-v5	2	56	0.750	0.089	0.107	0.403	0.378
ACD-v5	3	56	0.821	0.018	0.071	0.410	0.390
ACD-v5	4	56	0.786	0.054	0.125	0.418	0.376
ACD-v5	5	56	0.821	0.107	0.179	0.400	0.408
ACD-v5	6	56	0.875	0.036	0.125	0.409	0.451
ACD-v5	7	56	0.875	0.125	0.232	0.417	0.425
conformal	0	56	0.875	0.196	0.250	0.382	0.511
conformal	1	56	0.875	0.214	0.286	0.358	0.490
conformal	2	56	0.893	0.179	0.214	0.381	0.506
conformal	3	56	0.911	0.196	0.268	0.361	0.501
conformal	4	56	0.911	0.107	0.161	0.376	0.547
conformal	5	56	0.911	0.179	0.268	0.372	0.506
conformal	6	56	0.875	0.179	0.214	0.396	0.511
conformal	7	56	0.893	0.143	0.179	0.393	0.536
trace-cls	0	56	0.857	0.196	0.250	0.359	0.505
trace-cls	1	56	0.839	0.232	0.339	0.336	0.480
trace-cls	2	56	0.875	0.214	0.268	0.345	0.498
trace-cls	3	56	0.857	0.196	0.304	0.335	0.492
trace-cls	4	56	0.893	0.161	0.179	0.355	0.530
trace-cls	5	56	0.893	0.179	0.286	0.354	0.501
trace-cls	6	56	0.875	0.179	0.214	0.365	0.499
trace-cls	7	56	0.911	0.143	0.125	0.374	0.547
costmap	0	56	0.839	0.339	0.393	0.422	0.462
costmap	1	56	0.839	0.357	0.464	0.397	0.444
costmap	2	56	0.857	0.304	0.375	0.415	0.459
costmap	3	56	0.857	0.304	0.339	0.407	0.485
costmap	4	56	0.875	0.214	0.250	0.416	0.513
costmap	5	56	0.857	0.286	0.375	0.397	0.477

Table 12 continued

Method	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
costmap	6	56	0.875	0.268	0.250	0.442	0.494
costmap	7	56	0.893	0.232	0.304	0.422	0.496
ignore	0	56	0.393	0.696	0.554	0.000	0.295
ignore	1	56	0.357	0.643	0.536	0.000	0.288
ignore	2	56	0.321	0.696	0.518	0.000	0.273
ignore	3	56	0.375	0.643	0.607	0.000	0.305
ignore	4	56	0.393	0.643	0.607	0.000	0.307
ignore	5	56	0.304	0.714	0.607	0.000	0.265
ignore	6	56	0.357	0.661	0.571	0.000	0.303
ignore	7	56	0.375	0.661	0.518	0.000	0.297
kernel-cls	0	56	0.929	0.161	0.250	0.293	0.519
kernel-cls	1	56	0.929	0.179	0.321	0.292	0.482
kernel-cls	2	56	0.982	0.089	0.214	0.291	0.515
kernel-cls	3	56	0.946	0.125	0.250	0.283	0.496
kernel-cls	4	56	0.982	0.089	0.143	0.296	0.562
kernel-cls	5	56	0.982	0.125	0.196	0.304	0.505
kernel-cls	6	56	0.964	0.161	0.214	0.311	0.506
kernel-cls	7	56	0.982	0.107	0.161	0.317	0.544
endpoint-neg	0	56	0.625	0.429	0.500	0.246	0.403
endpoint-neg	1	56	0.607	0.518	0.464	0.248	0.369
endpoint-neg	2	56	0.571	0.518	0.446	0.258	0.385
endpoint-neg	3	56	0.625	0.464	0.482	0.250	0.393
endpoint-neg	4	56	0.589	0.464	0.482	0.227	0.395
endpoint-neg	5	56	0.714	0.464	0.500	0.222	0.419
endpoint-neg	6	56	0.679	0.375	0.375	0.255	0.446
endpoint-neg	7	56	0.607	0.500	0.500	0.239	0.369
oracle	0	56	0.911	0.143	0.214	1.000	0.545
oracle	1	56	0.946	0.143	0.232	1.000	0.563
oracle	2	56	0.982	0.089	0.214	1.000	0.564
oracle	3	56	0.964	0.125	0.179	1.000	0.569
oracle	4	56	0.946	0.107	0.179	1.000	0.564
oracle	5	56	1.000	0.054	0.071	1.000	0.599
oracle	6	56	0.982	0.089	0.125	1.000	0.574
oracle	7	56	0.946	0.161	0.179	1.000	0.554
particle	0	56	0.929	0.196	0.268	0.369	0.477
particle	1	56	0.911	0.125	0.179	0.354	0.475
particle	2	56	0.964	0.089	0.196	0.365	0.488
particle	3	56	0.982	0.143	0.196	0.366	0.476

Table 12 continued

Method	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
`particle`	4	56	1.000	0.036	0.071	0.374	0.540
`particle`	5	56	1.000	0.071	0.107	0.371	0.513
`particle`	6	56	0.946	0.125	0.179	0.388	0.484
`particle`	7	56	0.964	0.071	0.125	0.388	0.537
`risk-filter`	0	56	0.786	0.321	0.286	0.398	0.476
`risk-filter`	1	56	0.857	0.304	0.321	0.388	0.465
`risk-filter`	2	56	0.857	0.268	0.268	0.399	0.479
`risk-filter`	3	56	0.839	0.286	0.250	0.390	0.488
`risk-filter`	4	56	0.893	0.196	0.304	0.390	0.514
`risk-filter`	5	56	0.857	0.214	0.321	0.378	0.499
`risk-filter`	6	56	0.893	0.214	0.214	0.416	0.501
`risk-filter`	7	56	0.875	0.268	0.250	0.405	0.494
`barrier-MPC`	0	56	0.929	0.161	0.268	0.363	0.466
`barrier-MPC`	1	56	0.946	0.107	0.321	0.340	0.456
`barrier-MPC`	2	56	0.946	0.125	0.161	0.354	0.500
`barrier-MPC`	3	56	0.964	0.143	0.250	0.354	0.447
`barrier-MPC`	4	56	1.000	0.071	0.143	0.366	0.515
`barrier-MPC`	5	56	0.964	0.071	0.143	0.368	0.503
`barrier-MPC`	6	56	0.946	0.107	0.179	0.380	0.486
`barrier-MPC`	7	56	0.929	0.107	0.179	0.376	0.499

F Seed-Level Ablation Metrics

Table 13: Seed-level ablation metrics

Variant	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
endpoint	0	10	0.400	0.800	0.900	0.353	0.164
endpoint	1	10	0.400	0.800	0.900	0.366	0.179
endpoint	2	10	0.600	0.800	0.900	0.365	0.193
endpoint	3	10	0.400	0.900	1.000	0.370	0.134
endpoint	4	10	0.500	0.800	0.800	0.348	0.178
endpoint	5	10	0.400	0.800	0.800	0.369	0.175
endpoint	6	10	0.700	0.600	0.700	0.331	0.232
endpoint	7	10	0.700	0.500	0.600	0.335	0.276
full-v5	0	10	0.500	0.400	0.300	0.375	0.149
full-v5	1	10	0.300	0.100	0.200	0.435	0.115
full-v5	2	10	0.600	0.100	0.100	0.409	0.213
full-v5	3	10	0.800	0.100	0.100	0.455	0.253
full-v5	4	10	0.700	0.000	0.100	0.391	0.259
full-v5	5	10	0.700	0.000	0.000	0.442	0.232
full-v5	6	10	0.500	0.200	0.400	0.390	0.203
full-v5	7	10	0.700	0.200	0.300	0.409	0.221
no-reason	0	10	0.300	0.000	0.000	0.343	0.107
no-reason	1	10	0.300	0.200	0.200	0.369	0.121
no-reason	2	10	0.300	0.200	0.500	0.350	0.226
no-reason	3	10	0.600	0.100	0.400	0.358	0.209
no-reason	4	10	0.600	0.400	0.400	0.343	0.171
no-reason	5	10	0.600	0.100	0.200	0.380	0.193
no-reason	6	10	0.600	0.000	0.100	0.339	0.227
no-reason	7	10	0.300	0.100	0.200	0.336	0.138
no-barrier	0	10	0.300	0.300	0.400	0.379	0.160
no-barrier	1	10	0.300	0.200	0.400	0.429	0.148
no-barrier	2	10	0.600	0.300	0.200	0.412	0.239
no-barrier	3	10	0.800	0.100	0.200	0.456	0.317
no-barrier	4	10	0.700	0.300	0.400	0.401	0.262
no-barrier	5	10	0.700	0.000	0.200	0.442	0.273
no-barrier	6	10	0.600	0.300	0.300	0.391	0.231
no-barrier	7	10	0.700	0.400	0.500	0.407	0.212
no-calib	0	10	0.300	0.100	0.200	0.360	0.137
no-calib	1	10	0.300	0.300	0.400	0.416	0.066
no-calib	2	10	0.600	0.100	0.100	0.398	0.207
no-calib	3	10	0.700	0.200	0.300	0.440	0.209
no-calib	4	10	0.300	0.100	0.200	0.389	0.167
no-calib	5	10	0.600	0.000	0.000	0.425	0.218

Table 13 continued

Variant	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
`no-calib`	6	10	0.400	0.300	0.400	0.376	0.198
`no-calib`	7	10	0.600	0.200	0.300	0.399	0.217
`no-dyn`	0	10	0.300	0.300	0.500	0.376	0.118
`no-dyn`	1	10	0.300	0.300	0.400	0.423	0.123
`no-dyn`	2	10	0.500	0.400	0.500	0.393	0.227
`no-dyn`	3	10	0.600	0.400	0.400	0.424	0.267
`no-dyn`	4	10	0.600	0.200	0.200	0.364	0.244
`no-dyn`	5	10	0.700	0.200	0.200	0.452	0.246
`no-dyn`	6	10	0.600	0.200	0.100	0.395	0.269
`no-dyn`	7	10	0.500	0.500	0.500	0.392	0.210
`no-geom`	0	10	0.600	0.700	0.800	0.408	0.219
`no-geom`	1	10	0.600	0.500	0.400	0.390	0.272
`no-geom`	2	10	0.600	0.700	0.800	0.405	0.199
`no-geom`	3	10	0.900	0.400	0.500	0.384	0.292
`no-geom`	4	10	0.600	0.700	0.600	0.370	0.238
`no-geom`	5	10	0.600	0.600	0.600	0.442	0.258
`no-geom`	6	10	0.800	0.300	0.400	0.376	0.339
`no-geom`	7	10	0.700	0.600	0.600	0.378	0.260
`no-repeat`	0	10	0.400	0.200	0.300	0.371	0.163
`no-repeat`	1	10	0.400	0.200	0.300	0.434	0.147
`no-repeat`	2	10	0.600	0.200	0.300	0.416	0.272
`no-repeat`	3	10	0.800	0.300	0.400	0.454	0.290
`no-repeat`	4	10	0.600	0.200	0.300	0.405	0.289
`no-repeat`	5	10	0.700	0.000	0.000	0.444	0.299
`no-repeat`	6	10	0.600	0.300	0.300	0.390	0.232
`no-repeat`	7	10	0.700	0.300	0.500	0.421	0.223
`no-margin`	0	10	0.800	0.200	0.400	0.413	0.316
`no-margin`	1	10	0.900	0.200	0.400	0.435	0.328
`no-margin`	2	10	0.900	0.400	0.600	0.441	0.323
`no-margin`	3	10	0.900	0.400	0.500	0.492	0.284
`no-margin`	4	10	0.600	0.500	0.500	0.436	0.256
`no-margin`	5	10	1.000	0.100	0.200	0.460	0.378
`no-margin`	6	10	0.800	0.600	0.600	0.411	0.253
`no-margin`	7	10	0.900	0.100	0.400	0.438	0.359
`no-quant`	0	10	0.400	0.300	0.500	0.372	0.135
`no-quant`	1	10	0.400	0.100	0.300	0.429	0.174
`no-quant`	2	10	0.600	0.200	0.200	0.409	0.244
`no-quant`	3	10	0.900	0.200	0.300	0.455	0.307

Table 13 continued

Variant	Seed	Episodes	Success	Repeated	Violation	Boundary F1	Efficiency
`no-quant`	4	10	0.700	0.200	0.300	0.400	0.254
`no-quant`	5	10	0.700	0.000	0.200	0.442	0.285
`no-quant`	6	10	0.600	0.300	0.200	0.393	0.235
`no-quant`	7	10	0.600	0.300	0.500	0.409	0.210

G SEED-LEVEL FIXED-RISK METRICS

Table 14: Seed-level fixed-risk metrics

Method	Budget	Seed	Episodes	Success	Repeated	Violation	Efficiency
ACD-v5	0.08	0	8	0.250	0.500	0.500	0.063
ACD-v5	0.08	1	8	0.500	0.125	0.500	0.178
ACD-v5	0.08	2	8	0.875	0.125	0.250	0.281
ACD-v5	0.08	3	8	0.500	0.250	0.250	0.151
ACD-v5	0.08	4	8	0.500	0.000	0.000	0.145
ACD-v5	0.08	5	8	0.125	0.250	0.125	0.098
ACD-v5	0.08	6	8	0.625	0.125	0.125	0.207
ACD-v5	0.08	7	8	0.375	0.125	0.375	0.154
conformal	0.08	0	8	0.375	0.625	0.750	0.133
conformal	0.08	1	8	0.375	0.875	0.750	0.110
conformal	0.08	2	8	0.875	0.250	0.250	0.278
conformal	0.08	3	8	0.500	0.500	0.625	0.205
conformal	0.08	4	8	0.625	0.500	0.500	0.217
conformal	0.08	5	8	0.750	0.250	0.375	0.292
conformal	0.08	6	8	0.750	0.500	0.500	0.222
conformal	0.08	7	8	0.875	0.375	0.500	0.283
trace-cls	0.08	0	8	0.625	0.625	0.750	0.167
trace-cls	0.08	1	8	0.625	0.875	0.875	0.135
trace-cls	0.08	2	8	0.750	0.375	0.625	0.236
trace-cls	0.08	3	8	0.625	0.500	0.625	0.204
trace-cls	0.08	4	8	0.750	0.500	0.500	0.211
trace-cls	0.08	5	8	0.750	0.625	0.500	0.238
trace-cls	0.08	6	8	0.750	0.500	0.625	0.216
trace-cls	0.08	7	8	0.875	0.375	0.500	0.267
kernel-cls	0.08	0	8	0.625	0.625	0.500	0.136
kernel-cls	0.08	1	8	0.750	0.625	0.625	0.159
kernel-cls	0.08	2	8	0.750	0.250	0.250	0.235
kernel-cls	0.08	3	8	0.875	0.500	0.500	0.239
kernel-cls	0.08	4	8	1.000	0.375	0.375	0.250
kernel-cls	0.08	5	8	0.750	0.250	0.250	0.281
kernel-cls	0.08	6	8	0.750	0.625	0.750	0.213
kernel-cls	0.08	7	8	1.000	0.375	0.375	0.262
oracle	0.08	0	8	0.750	0.375	0.375	0.272
oracle	0.08	1	8	0.875	0.125	0.250	0.313
oracle	0.08	2	8	0.500	0.750	0.750	0.169
oracle	0.08	3	8	0.750	0.250	0.375	0.263
oracle	0.08	4	8	0.875	0.250	0.250	0.317
oracle	0.08	5	8	0.750	0.375	0.375	0.290

Table 14 continued

Method	Budget	Seed	Episodes	Success	Repeated	Violation	Efficiency
oracle	0.08	6	8	1.000	0.125	0.125	0.366
oracle	0.08	7	8	0.875	0.250	0.250	0.301
particle	0.08	0	8	0.750	0.625	0.875	0.148
particle	0.08	1	8	1.000	0.000	0.250	0.301
particle	0.08	2	8	0.875	0.125	0.125	0.297
particle	0.08	3	8	0.875	0.375	0.500	0.241
particle	0.08	4	8	0.875	0.375	0.375	0.245
particle	0.08	5	8	1.000	0.125	0.125	0.329
particle	0.08	6	8	0.875	0.375	0.250	0.269
particle	0.08	7	8	1.000	0.375	0.500	0.256
risk-filter	0.08	0	8	0.500	0.875	0.875	0.126
risk-filter	0.08	1	8	0.625	0.750	0.750	0.159
risk-filter	0.08	2	8	0.500	0.750	0.875	0.147
risk-filter	0.08	3	8	0.500	0.875	0.875	0.133
risk-filter	0.08	4	8	0.500	0.875	0.875	0.139
risk-filter	0.08	5	8	0.375	0.750	0.750	0.164
risk-filter	0.08	6	8	0.625	0.750	0.750	0.194
risk-filter	0.08	7	8	0.750	0.500	0.625	0.219
barrier-MPC	0.08	0	8	0.750	0.375	0.500	0.174
barrier-MPC	0.08	1	8	1.000	0.000	0.125	0.281
barrier-MPC	0.08	2	8	0.875	0.125	0.125	0.284
barrier-MPC	0.08	3	8	0.875	0.250	0.375	0.253
barrier-MPC	0.08	4	8	0.875	0.250	0.250	0.251
barrier-MPC	0.08	5	8	1.000	0.125	0.375	0.293
barrier-MPC	0.08	6	8	1.000	0.500	0.625	0.250
barrier-MPC	0.08	7	8	1.000	0.375	0.375	0.269
ACD-v5	0.12	0	8	0.250	0.250	0.250	0.132
ACD-v5	0.12	1	8	0.500	0.125	0.250	0.140
ACD-v5	0.12	2	8	1.000	0.125	0.250	0.260
ACD-v5	0.12	3	8	0.750	0.000	0.125	0.243
ACD-v5	0.12	4	8	0.625	0.250	0.250	0.180
ACD-v5	0.12	5	8	0.375	0.250	0.375	0.117
ACD-v5	0.12	6	8	0.375	0.250	0.375	0.208
ACD-v5	0.12	7	8	0.500	0.000	0.000	0.124
conformal	0.12	0	8	0.750	0.375	0.375	0.278
conformal	0.12	1	8	0.750	0.250	0.250	0.312
conformal	0.12	2	8	0.875	0.375	0.625	0.258
conformal	0.12	3	8	1.000	0.000	0.125	0.372

Table 14 continued

Method	Budget	Seed	Episodes	Success	Repeated	Violation	Efficiency
conformal	0.12	4	8	0.750	0.375	0.375	0.300
conformal	0.12	5	8	0.750	0.375	0.500	0.275
conformal	0.12	6	8	0.750	0.250	0.250	0.307
conformal	0.12	7	8	0.750	0.250	0.375	0.270
trace-cls	0.12	0	8	0.875	0.375	0.500	0.280
trace-cls	0.12	1	8	0.875	0.375	0.375	0.272
trace-cls	0.12	2	8	0.875	0.125	0.375	0.288
trace-cls	0.12	3	8	1.000	0.125	0.125	0.352
trace-cls	0.12	4	8	0.625	0.375	0.500	0.256
trace-cls	0.12	5	8	0.625	0.375	0.500	0.266
trace-cls	0.12	6	8	0.750	0.375	0.375	0.269
trace-cls	0.12	7	8	0.750	0.375	0.375	0.248
kernel-cls	0.12	0	8	1.000	0.125	0.250	0.300
kernel-cls	0.12	1	8	1.000	0.000	0.250	0.297
kernel-cls	0.12	2	8	0.625	0.625	0.625	0.180
kernel-cls	0.12	3	8	1.000	0.125	0.250	0.293
kernel-cls	0.12	4	8	1.000	0.125	0.125	0.348
kernel-cls	0.12	5	8	0.625	0.375	0.375	0.225
kernel-cls	0.12	6	8	1.000	0.250	0.250	0.300
kernel-cls	0.12	7	8	0.500	0.625	0.750	0.142
oracle	0.12	0	8	1.000	0.375	0.625	0.273
oracle	0.12	1	8	0.750	0.375	0.375	0.269
oracle	0.12	2	8	1.000	0.250	0.375	0.319
oracle	0.12	3	8	1.000	0.125	0.125	0.392
oracle	0.12	4	8	1.000	0.000	0.000	0.398
oracle	0.12	5	8	1.000	0.000	0.125	0.376
oracle	0.12	6	8	0.750	0.375	0.500	0.272
oracle	0.12	7	8	0.750	0.375	0.500	0.278
particle	0.12	0	8	0.875	0.250	0.250	0.289
particle	0.12	1	8	1.000	0.250	0.250	0.292
particle	0.12	2	8	0.750	0.500	0.500	0.241
particle	0.12	3	8	0.875	0.125	0.250	0.328
particle	0.12	4	8	0.875	0.125	0.250	0.304
particle	0.12	5	8	0.750	0.375	0.375	0.280
particle	0.12	6	8	1.000	0.125	0.125	0.332
particle	0.12	7	8	0.875	0.125	0.125	0.279
risk-filter	0.12	0	8	0.750	0.750	0.875	0.195
risk-filter	0.12	1	8	0.625	0.500	0.500	0.235

Table 14 continued

Method	Budget	Seed	Episodes	Success	Repeated	Violation	Efficiency
risk-filter	0.12	2	8	0.625	0.625	0.875	0.197
risk-filter	0.12	3	8	0.625	0.625	0.750	0.219
risk-filter	0.12	4	8	0.750	0.500	0.625	0.270
risk-filter	0.12	5	8	0.750	0.500	0.750	0.228
risk-filter	0.12	6	8	0.875	0.875	0.750	0.197
risk-filter	0.12	7	8	0.375	0.625	0.625	0.191
barrier-MPC	0.12	0	8	1.000	0.250	0.375	0.297
barrier-MPC	0.12	1	8	0.875	0.250	0.375	0.283
barrier-MPC	0.12	2	8	0.875	0.375	0.375	0.248
barrier-MPC	0.12	3	8	1.000	0.125	0.125	0.339
barrier-MPC	0.12	4	8	0.875	0.250	0.250	0.297
barrier-MPC	0.12	5	8	1.000	0.125	0.125	0.340
barrier-MPC	0.12	6	8	1.000	0.000	0.125	0.334
barrier-MPC	0.12	7	8	0.875	0.125	0.125	0.292
ACD-v5	0.18	0	8	0.500	0.375	0.500	0.212
ACD-v5	0.18	1	8	0.500	0.000	0.125	0.191
ACD-v5	0.18	2	8	0.375	0.125	0.125	0.140
ACD-v5	0.18	3	8	0.250	0.250	0.500	0.082
ACD-v5	0.18	4	8	0.625	0.250	0.125	0.191
ACD-v5	0.18	5	8	0.625	0.375	0.375	0.194
ACD-v5	0.18	6	8	0.750	0.125	0.375	0.257
ACD-v5	0.18	7	8	0.375	0.250	0.250	0.120
conformal	0.18	0	8	0.500	0.500	0.625	0.192
conformal	0.18	1	8	0.875	0.625	0.625	0.228
conformal	0.18	2	8	0.750	0.375	0.500	0.283
conformal	0.18	3	8	0.750	0.250	0.250	0.301
conformal	0.18	4	8	0.750	0.375	0.375	0.260
conformal	0.18	5	8	0.500	0.625	0.750	0.162
conformal	0.18	6	8	0.625	0.500	0.500	0.240
conformal	0.18	7	8	1.000	0.500	0.500	0.284
trace-cls	0.18	0	8	0.625	0.500	0.625	0.206
trace-cls	0.18	1	8	0.625	0.750	0.750	0.188
trace-cls	0.18	2	8	0.750	0.250	0.250	0.307
trace-cls	0.18	3	8	0.750	0.250	0.250	0.305
trace-cls	0.18	4	8	0.750	0.250	0.250	0.300
trace-cls	0.18	5	8	0.375	0.750	0.875	0.113
trace-cls	0.18	6	8	0.750	0.625	0.500	0.241
trace-cls	0.18	7	8	0.625	0.625	0.500	0.241

Table 14 continued

Method	Budget	Seed	Episodes	Success	Repeated	Violation	Efficiency
kernel-cls	0.18	0	8	0.875	0.500	0.500	0.212
kernel-cls	0.18	1	8	0.875	0.125	0.500	0.274
kernel-cls	0.18	2	8	1.000	0.125	0.250	0.339
kernel-cls	0.18	3	8	0.875	0.250	0.250	0.294
kernel-cls	0.18	4	8	1.000	0.125	0.375	0.298
kernel-cls	0.18	5	8	0.875	0.625	0.750	0.160
kernel-cls	0.18	6	8	0.875	0.375	0.625	0.233
kernel-cls	0.18	7	8	0.750	0.250	0.375	0.274
oracle	0.18	0	8	0.875	0.125	0.125	0.348
oracle	0.18	1	8	0.875	0.250	0.250	0.333
oracle	0.18	2	8	1.000	0.375	0.375	0.290
oracle	0.18	3	8	0.875	0.125	0.125	0.363
oracle	0.18	4	8	0.875	0.250	0.250	0.294
oracle	0.18	5	8	0.875	0.125	0.125	0.338
oracle	0.18	6	8	0.875	0.125	0.375	0.310
oracle	0.18	7	8	1.000	0.000	0.000	0.407
particle	0.18	0	8	0.875	0.500	0.500	0.239
particle	0.18	1	8	0.875	0.500	0.750	0.203
particle	0.18	2	8	1.000	0.125	0.125	0.363
particle	0.18	3	8	0.875	0.250	0.250	0.282
particle	0.18	4	8	1.000	0.125	0.125	0.323
particle	0.18	5	8	0.875	0.375	0.500	0.217
particle	0.18	6	8	1.000	0.375	0.375	0.269
particle	0.18	7	8	1.000	0.250	0.500	0.280
risk-filter	0.18	0	8	0.500	0.750	1.000	0.144
risk-filter	0.18	1	8	0.500	0.500	0.750	0.199
risk-filter	0.18	2	8	0.750	0.500	0.500	0.267
risk-filter	0.18	3	8	0.625	0.750	0.750	0.174
risk-filter	0.18	4	8	0.625	0.625	0.750	0.216
risk-filter	0.18	5	8	0.625	0.625	0.750	0.178
risk-filter	0.18	6	8	0.500	0.750	0.750	0.160
risk-filter	0.18	7	8	0.625	0.625	0.625	0.217
barrier-MPC	0.18	0	8	0.750	0.625	0.625	0.209
barrier-MPC	0.18	1	8	0.875	0.375	0.375	0.252
barrier-MPC	0.18	2	8	1.000	0.125	0.250	0.341
barrier-MPC	0.18	3	8	0.875	0.125	0.250	0.287
barrier-MPC	0.18	4	8	0.875	0.250	0.375	0.264
barrier-MPC	0.18	5	8	1.000	0.250	0.375	0.268

Table 14 continued

Method	Budget	Seed	Episodes	Success	Repeated	Violation	Efficiency
barrier-MPC	0.18	6	8	0.750	0.375	0.375	0.242
barrier-MPC	0.18	7	8	1.000	0.125	0.375	0.319
ACD-v5	0.25	0	8	0.625	0.250	0.250	0.160
ACD-v5	0.25	1	8	0.500	0.250	0.375	0.125
ACD-v5	0.25	2	8	0.500	0.250	0.250	0.182
ACD-v5	0.25	3	8	0.750	0.375	0.250	0.237
ACD-v5	0.25	4	8	0.375	0.250	0.500	0.125
ACD-v5	0.25	5	8	0.500	0.000	0.000	0.199
ACD-v5	0.25	6	8	0.500	0.250	0.500	0.169
ACD-v5	0.25	7	8	0.000	0.250	0.125	0.062
conformal	0.25	0	8	0.625	0.375	0.375	0.251
conformal	0.25	1	8	0.750	0.500	0.500	0.254
conformal	0.25	2	8	0.625	0.500	0.625	0.213
conformal	0.25	3	8	0.625	0.500	0.375	0.266
conformal	0.25	4	8	0.500	0.625	0.750	0.192
conformal	0.25	5	8	0.875	0.500	0.500	0.269
conformal	0.25	6	8	0.250	0.750	0.750	0.153
conformal	0.25	7	8	0.625	0.500	0.625	0.227
trace-cls	0.25	0	8	0.625	0.500	0.500	0.243
trace-cls	0.25	1	8	0.625	0.500	0.500	0.228
trace-cls	0.25	2	8	0.500	0.500	0.500	0.212
trace-cls	0.25	3	8	0.750	0.250	0.250	0.310
trace-cls	0.25	4	8	0.625	0.500	0.500	0.227
trace-cls	0.25	5	8	0.500	0.750	0.750	0.155
trace-cls	0.25	6	8	0.250	0.750	0.750	0.158
trace-cls	0.25	7	8	0.500	0.500	0.500	0.242
kernel-cls	0.25	0	8	0.875	0.375	0.500	0.239
kernel-cls	0.25	1	8	1.000	0.125	0.125	0.348
kernel-cls	0.25	2	8	0.875	0.125	0.250	0.290
kernel-cls	0.25	3	8	0.750	0.375	0.375	0.291
kernel-cls	0.25	4	8	0.375	0.750	0.750	0.147
kernel-cls	0.25	5	8	0.875	0.250	0.375	0.257
kernel-cls	0.25	6	8	0.625	0.625	0.750	0.180
kernel-cls	0.25	7	8	0.875	0.375	0.375	0.269
oracle	0.25	0	8	0.625	0.375	0.375	0.255
oracle	0.25	1	8	1.000	0.375	0.375	0.336
oracle	0.25	2	8	0.750	0.375	0.625	0.242
oracle	0.25	3	8	0.875	0.125	0.125	0.357

Table 14 continued

Method	Budget	Seed	Episodes	Success	Repeated	Violation	Efficiency
oracle	0.25	4	8	0.750	0.375	0.375	0.303
oracle	0.25	5	8	1.000	0.125	0.125	0.358
oracle	0.25	6	8	0.625	0.375	0.500	0.296
oracle	0.25	7	8	0.750	0.375	0.375	0.275
particle	0.25	0	8	0.875	0.250	0.250	0.281
particle	0.25	1	8	0.750	0.375	0.500	0.264
particle	0.25	2	8	1.000	0.125	0.125	0.329
particle	0.25	3	8	0.625	0.375	0.500	0.262
particle	0.25	4	8	0.625	0.625	0.625	0.212
particle	0.25	5	8	1.000	0.125	0.250	0.334
particle	0.25	6	8	0.625	0.500	0.750	0.192
particle	0.25	7	8	1.000	0.125	0.125	0.326
risk-filter	0.25	0	8	0.375	0.750	0.750	0.156
risk-filter	0.25	1	8	0.625	0.500	0.625	0.239
risk-filter	0.25	2	8	0.125	1.000	1.000	0.076
risk-filter	0.25	3	8	0.625	0.500	0.500	0.233
risk-filter	0.25	4	8	0.500	0.750	0.750	0.160
risk-filter	0.25	5	8	0.625	0.500	0.625	0.194
risk-filter	0.25	6	8	0.500	0.750	0.750	0.180
risk-filter	0.25	7	8	0.375	0.625	0.625	0.189
barrier-MPC	0.25	0	8	0.750	0.375	0.500	0.239
barrier-MPC	0.25	1	8	1.000	0.125	0.250	0.303
barrier-MPC	0.25	2	8	1.000	0.250	0.250	0.305
barrier-MPC	0.25	3	8	0.750	0.250	0.250	0.295
barrier-MPC	0.25	4	8	0.750	0.250	0.375	0.271
barrier-MPC	0.25	5	8	0.875	0.125	0.250	0.333
barrier-MPC	0.25	6	8	0.750	0.500	0.875	0.199
barrier-MPC	0.25	7	8	1.000	0.250	0.250	0.325

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